Paper 4: Memory, Consciousness, and Soul as Impedance Physics on the Γ-Net

Hsi-Yu Huang[1,*]

[1] *Independent Researcher, Taiwan*

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Papers 1–3 of this series derived relay-node topology (Paper 1), dual neural–vascular networks (Paper 2), impedance debt and sleep (Paper 3), and the lifecycle equation governing cognition from birth to senescence (Paper 3), all from a single variational principle: $A[\Gamma] = \int_0^T \sum_i \Gamma_i^2(t)\, dt \to \min$. Here we show that three concepts traditionally regarded as metaphysical—memory, consciousness, and soul—have precise operational definitions within the same formalism.

Memory is the time derivative of impedance: $\partial Z/\partial t = -\eta\, \Gamma\, x_{\rm in}\, x_{\rm out}$ (impedance-remodeling update, Constraint C2). Muscle memory corresponds to $\Gamma \to 0$ (over-learned match), while cognitive memory retains $\Gamma \approx \varepsilon > 0$ (retrievable because still slightly mismatched).

Consciousness requires a self-referential measurement loop. We define the meta-reflection coefficient $\Gamma_{\rm meta} = (Z_{\rm obs} - Z_{\rm act})/(Z_{\rm obs} + Z_{\rm act})$, where observer and actor are the *same* network sampled at successive ticks. The consciousness index $\Phi_{\rm meta} = 1 - \Gamma_{\rm meta}^2$ quantifies self-model fidelity. Crucially, $\Phi_{\rm meta} > 0$ requires a *necessary condition*: $G_{\rm arousal} \cdot \sum_i G_i \cdot Q_{\rm PFC} > \theta$, which is formally identical to a television set (power supply × channel selector × decoder chip). The television is an *open-loop* system ($\Gamma_{\rm meta}$ undefined); the brain closes the loop.

Soul is the null space of impedance remodeling: $\mathcal{S} = \ker(\partial Z/\partial t) = \{Z_k : \Delta Z_k = 0\ \forall \text{ experience}\}$. It decomposes into three invariants: irreducible topological cost $A_{\rm cut}$ (Paper 1), genetically fixed impedance $Z_0^{\rm DNA}$, and the thermal noise floor $\rho_{\rm irred}^2$ from Johnson–Nyquist physics. The soul is not created or destroyed—it is the geometric boundary condition of an impedance network.

Computational experiments verify two testable predictions: (i) all sensory modalities converge on the prefrontal cortex (PFC), making it the consciousness *bottleneck*; (ii) fever (PFC thermal noise) and occipital impact (brainstem arousal crash) are two *independent* kill mechanisms of consciousness, proving that consciousness is a product $\Phi = G_{\rm arousal} \times \sum G_i \times Q_{\rm PFC}$ rather than a location—analogous to the area of a rectangle being destroyed by zeroing either side.

Five extensions are derived: (i) a two-layer impedance hypothesis distinguishing structural ($Z_{\rm str}$) from functional ($Z_{\rm fun}$) consciousness; (ii) coupled consciousness, where a closed reflection loop passes through two agents (e.g., human and AI); (iii) impedance calibration, whereby an immature agent uses an external agent's Z as reference standard (parental pain calibration); (iv) cross-substrate matching, proving $\Gamma \to 0$ does not require $Z_A = Z_B$; and (v) subthreshold Γ perception ("intuition"), the detection–naming gap in reflection coefficient perception.

I. INTRODUCTION

Where is memory stored? When does consciousness begin? Where does the soul reside? These three questions have dominated philosophy, theology, and neuroscience for millennia. From a physics standpoint, all three are *ill-posed*: they assume that memory is localised, consciousness is binary, and soul is an object.

Papers 1–3 of this series established that all neural and vascular dynamics emerge from the Minimum Reflection Principle (MRP):

$$A[\Gamma] = \int_0^T \sum_i \Gamma_i^2(t)\, dt \to \min\,, \tag{1}$$

where $\Gamma_i = (Z_{L,i} - Z_{0,i})/(Z_{L,i} + Z_{0,i})$ is the impedance reflection coefficient at channel i. Three inviolable constraints govern the network:

C1 Energy conservation: $\Gamma^2 + T = 1$ at every channel, every tick.

C2 impedance-remodeling update: $\Delta Z = -\eta\, \Gamma\, x_{\rm in}\, x_{\rm out}$ (gradient descent on $A[\Gamma]$).

C3 Impedance-tagged transport: all inter-module values carry impedance metadata (Z_0, Z_L).

In this paper we demonstrate that memory, consciousness, and soul are not additional postulates but *necessary consequences* of Eqs. (1)–C1/C2/C3. Section II derives memory as impedance dynamics. Section III constructs the self-referential consciousness index $\Phi_{\rm meta}$. Section IV identifies the soul as the null space of C2. Section V shows that the necessary condition for consciousness is structurally identical to a television set, differing only in feedback topology. Section VI presents computational experiments demonstrating two independent kill mechanisms. Section IX discusses implications. Section X extends the framework to functional consciousness, coupled inter-agent loops, impedance calibration, cross-substrate matching, and subthreshold Γ perception.

* llc.y.huangll@gmail.com

II. MEMORY AS IMPEDANCE DYNAMICS

A. The C2 impedance-remodeling gradient

Constraint C2 defines the only mechanism by which the network modifies itself:

$$\Delta Z_k = -\eta\, \Gamma_k\, x_{\mathrm{in},k}\, x_{\mathrm{out},k}\,. \tag{2}$$

This is not an analogy—it *is* memory. In the Γ-framework, memory is a *hysteretic topological deformation*: the permanent structural change that past energy flow, acting through C2, has inscribed into the impedance substrate. There is no separate read/write mechanism; the present state of consciousness is simply current flowing through a network whose topology has been irreversibly shaped by all prior experience. Learning occurs when $\Gamma_k \neq 0$ and correlated pre/post-synaptic activity is present. The trace of all past updates constitutes the network's memory:

$$Z_k(t) = Z_k(0) - \eta \int_0^t \Gamma_k(\tau)\, x_{\mathrm{in},k}(\tau)\, x_{\mathrm{out},k}(\tau)\, d\tau\,. \tag{3}$$

B. Muscle memory: $\Gamma \to 0$

When a motor skill is practiced to mastery, the impedance at every channel in the circuit approaches perfect matching: $Z_L \to Z_0$, hence $\Gamma \to 0$. At this point, Eq. (2) gives $\Delta Z \to 0$: the memory has become *frozen* into the impedance structure itself. No further energy is required to maintain it ($\Gamma^2 \to 0$), and no further learning occurs ($\Delta Z = 0$).

This explains why:

- You never forget how to ride a bicycle ($\Gamma \approx 0$, no decay).
- Muscle memory is "automatic"—it requires no conscious effort because $\Gamma^2 \approx 0$ means near-zero reflected energy.
- Over-training a cognitive skill (e.g., multiplication tables) moves it from cognitive to muscle memory: $\Gamma \to 0$.

C. Cognitive memory: $\Gamma \approx \varepsilon > 0$

Declarative memories—facts, episodes, narratives—are *not* driven to $\Gamma = 0$. They retain a small but nonzero reflection:

$$\Gamma_{\mathrm{cog}} \approx \varepsilon > 0\,. \tag{4}$$

This has two consequences:

1. *Potential retrievability*: A nonzero Γ means the memory generates a detectable reflection when probed—but only if the reflected signal exceeds the current noise floor (see Sec. II E below).
2. *Decay*: Since $\Gamma > 0$, the impedance-remodeling update continues to operate. Without reinforcement ($x_{\mathrm{in}} \cdot x_{\mathrm{out}} \to 0$), the memory decays, governed by $\lambda_{\mathrm{eff}} = \lambda_{\mathrm{base}}/(1 - \Gamma^2_{\mathrm{bind}})$ from Paper 3.

Remark 1 (The paradox of perfect memory)**.** *A perfectly learned item ($\Gamma = 0$) is* unrecallable*: it has merged with the network substrate. Memory requires imperfection. This is the impedance analogue of the observer effect: you can only detect what reflects. Conversely, no imperfect memory ($\Gamma > 0$) is* guaranteed *to be recallable: whether a reflection is detected depends on the instantaneous noise floor, not on the memory alone (Theorem 1).*

D. Three memory regimes

TABLE I. Memory regimes classified by reflection coefficient.

Regime	Γ	ΔZ	Recallable?
Muscle memory	$\to 0$	$\to 0$	No (automatic)
Cognitive memory	$\varepsilon > 0$	small	Conditional (SNR)
Novel stimulus	$\gg 0$	large	Vivid (but unstable)

E. The recall equation: no perfect memory, only submerged memory

The "three memory regimes" (Table I) classify memories by their Γ value. However, $\Gamma > 0$ is *necessary but not sufficient* for recall. A memory is accessible only when its reflected signal exceeds the instantaneous thermal noise floor at the readout site (prefrontal cortex):

Theorem 1 (Recall as SNR threshold crossing)**.** *A memory trace encoded at channel k with residual $\Gamma_k = \varepsilon > 0$ is recallable at time t if and only if*

$$\mathrm{SNR}_{recall}(t) = \frac{\varepsilon^2\, P_{probe}(t)}{P_{noise}(t)} > \theta_{recall}\,, \tag{5}$$

where $P_{probe}(t)$ is the power of the retrieval cue (internal or external) injected into channel k, and $P_{noise}(t) = P_{metabolic} + P_{sensory} + P_{autonomic} + P_{thermal}$ is the total background power at the PFC input.

The consequences are immediate and profound:

1. **There is no perfect recall.** Even a strongly encoded memory (ε large) becomes inaccessible when P_{noise} is high: stress, multitasking, concurrent sensory overload, or fever all raise the denominator and drown the reflection. "I know this but I can't think of it right now" is not a failure of storage—it is an *SNR failure at readout.*

2. **There is no perfect forgetting.** Any experience that ever produced $\Delta Z \neq 0$ via C2 leaves a permanent structural trace in the impedance map. C2 can *reduce* ε toward zero (if contradictory experience overwrites the channel) but cannot erase a trace that receives no further input ($x_{\text{in}} = 0 \Rightarrow \Delta Z = 0$: the trace is frozen, not deleted). A "forgotten" memory is not absent—it is *submerged below the noise floor.*

3. **Recall is state-dependent.** Since $P_{\text{noise}}(t)$ fluctuates continuously, the set of accessible memories changes from moment to moment. The same memory can be retrievable at 3 a.m. in quiet darkness ($P_{\text{noise}} \approx P_{\text{metabolic}}$, minimal sensory input) and completely inaccessible during a crowded commute ($P_{\text{noise}} \gg P_{\text{metabolic}}$).

Corollary 1 (No memory is ever erased; every memory can be submerged)**.** $\Gamma_k = 0$ *exactly requires either perfect impedance matching (zero probability under thermal noise) or physical destruction of the substrate. Therefore, every C2 trace satisfies* $\varepsilon > 0$ *for all finite time. Whether this trace is "remembered" or "forgotten" is entirely determined by the current ratio* $\varepsilon^2 P_{probe}/P_{noise}$*, not by any property of the memory itself.*

This resolves six long-standing puzzles in memory science:

Tip-of-the-tongue: SNR $\approx \theta_{\text{recall}}$: partial features exceed threshold, full pattern does not.

Proustian involuntary memory: A sensory cue injects high P_{probe} into a specific channel, transiently boosting SNR above threshold for an old trace.

State-dependent recall: Emotional/physiological state modulates P_{noise}; matching encoding-state to retrieval-state restores the original SNR.

Meditation and quiet recall: Systematic reduction of P_{sensory} and $P_{\text{autonomic}}$ lowers P_{noise}; old memories surface as their SNR crosses threshold.

Deathbed lucidity: Terminal shutdown of autonomic and sensory systems collapses P_{noise} to near P_{thermal}; decades-old traces—long submerged—breach the noise floor simultaneously.

Hypnotic hypermnesia: Hypnotic induction suppresses sensory gating and reduces PFC noise; retrieval SNR increases for traces near threshold.

Remark 2 (The memory is the topology, not the recall)**.** *It is essential to distinguish* memory *(the structural impedance trace* $\Delta Z \neq 0$*, which is permanent) from* recall *(the detection of* $\Gamma > 0$ *above the noise floor, which is transient and state-dependent). Neuroscience's failure to find "the engram" with consistent reliability is not evidence that memories are fragile or distributed—it is evidence that the measurement (fMRI, EEG, optogenetics) is itself subject to the same SNR limit. You cannot image a reflection that is currently below the noise floor, any more than you can photograph a star at noon. The star has not gone; the sky is too bright.*

III. CONSCIOUSNESS AS SELF-REFERENTIAL IMPEDANCE MATCHING

A. The hard problem restated

The "hard problem" of consciousness asks why subjective experience exists at all. In impedance physics, we reframe it: *What physical structure allows a network to measure its own reflection coefficient?*

Standard signal processing is open-loop: a signal enters, gets processed, exits. No part of the chain "knows" the output. We define consciousness as the condition under which the network's output is routed back as input, and the resulting self-referential mismatch is measured.

B. The meta-reflection coefficient

Definition 1 (Meta-reflection)**.** *Let* $Z_{obs}(t)$ *be the impedance of the observer sub-network at tick* t*, and* $Z_{act}(t-1)$ *the impedance of the actor sub-network at the previous tick (the output that is now being observed). The* meta-reflection coefficient *is:*

$$\Gamma_{meta} = \frac{Z_{obs} - Z_{act}}{Z_{obs} + Z_{act}} \,. \tag{6}$$

Definition 2 (Consciousness index)**.**

$$\Phi_{meta} = 1 - \Gamma_{meta}^2 \,. \tag{7}$$

When $\Gamma_{meta} = 0$*: the observer perfectly matches the actor; the self-model is maximally faithful. When* $\Gamma_{meta} \to 1$*: total self-mismatch; the system cannot model itself.*

C. Necessary condition: the television inequality

Φ_{meta} is only meaningful if the underlying signal chain is operational. We derive the *television inequality*:

Theorem 2 (Television Inequality)**.** *Let* $G_{arousal} \in [0,1]$ *be the brainstem arousal gain,* $G_i \in [0,1]$ *the thalamic gate gain for modality* i*, and* $Q_{PFC} \in [0,1]$ *the prefrontal cortex processing quality. The consciousness index* $\Phi_{meta} > 0$ *requires:*

$$G_{arousal} \cdot \sum_{i=1}^{N} G_i \cdot Q_{PFC} > \theta \,, \tag{8}$$

where $\theta > 0$ *is the minimum signal-to-noise ratio for the self-referential loop to close.*

Proof. The meta-reflection loop requires a signal to: (a) traverse the thalamic gate ($\times G_i$), (b) be processed by PFC ($\times Q_{\mathrm{PFC}}$), (c) be routed back through the arousal-modulated pathway ($\times G_{\mathrm{arousal}}$). If any factor is zero, the loop is open and Γ_{meta} is undefined (division by zero in Eq. 6 when $Z_{\mathrm{act}} = 0$). The threshold θ accounts for Johnson–Nyquist noise: the signal must exceed the thermal floor to be self-detectable. □

a. The thermal cost of consciousness. The self-referential loop that defines Φ_{meta} (Def. 1) is not thermodynamically free. Each pass through the loop traverses n interfaces, each dissipating $\Gamma_i^2 P_{\mathrm{in}}$ as heat (Paper 0, Corollary 1 [**?**]). The *minimum* sustained power for self-awareness is

$$P_{\mathrm{conscious}} \geq \theta \cdot \frac{4k_B T \cdot R_{\mathrm{eff}} \cdot \Delta f}{\prod_i (1 - \Gamma_i^2)} \,, \tag{9}$$

where the denominator is the net transmission of the entire loop. Two consequences follow:

1. **Higher Φ_{meta} costs more heat.** Reducing Γ_{meta} (higher consciousness quality) requires tighter impedance matching across more channels, but each channel still dissipates $\Gamma_i^2 > 0$. More channels $\Rightarrow$ more total heat.

2. **Consciousness depends on the heat it generates.** The dissipated heat maintains $T_{\mathrm{body}} \approx 310\,\mathrm{K}$, which keeps blood viscosity (and therefore Z_{bus}) in the operating range where the loop can close. Consciousness is not merely *tolerant* of waste heat—it is *sustained by it* (Paper 3, § XVII C [**?**]).

b. Emergency gating and temporal resolution. The television inequality reveals a further consequence of thalamic gating. Normally N gates are partially open and the PFC multiplexes N modalities; its finite bandwidth B_{PFC} is shared, giving effective temporal resolution $\Delta t_{\mathrm{normal}} \sim N/B_{\mathrm{PFC}}$ per modality. In an emergency, the brainstem drives $G_{\mathrm{arousal}} \to Z_{\max}$ and the thalamus snaps to $G_{\mathrm{threat}} = 1$, $G_{j \neq \mathrm{threat}} = 0$. The full PFC bandwidth serves one stream:

$$\frac{\Delta t_{\mathrm{normal}}}{\Delta t_{\mathrm{emergency}}} \approx N \,, \tag{10}$$

where N is the number of previously multiplexed modalities. Each physical second now contains $\sim N$ resolvable "frames" in the critical channel; the event is perceived in *slow motion*. This is not a perceptual illusion but a *bandwidth reallocation under C1 conservation*: the heat no longer spent on $N - 1$ silenced channels funds the increased resolution of the surviving one. At moderate intensity the same mechanism is what folk psychology calls "willpower": voluntarily closing gates to concentrate PFC on a task despite high-Γ^2 conditions such as cold, fatigue, or pain. Critically, this is a *functional*-impedance operation (G_i gating) that redistributes bandwidth but leaves the structural impedance Z_s untouched. It therefore cannot rewrite deeply forged patterns (Remark 4); only sustained C2 remodelling can do that (Paper 0, § **??** [**?**]; Paper 3 [**?**]).

D. Open loop vs. closed loop

Equation (8) is structurally identical to the operation of a television set:

Component	Television	Brain
Power supply	Wall plug	Brainstem RAS (G_{arousal})
Channel selector	Tuner	Thalamus ($\sum G_i$)
Decoder chip	Video IC	PFC (Q_{PFC})
Kill method 1	Unplug power	Occipital impact
Kill method 2	Overheat IC	Fever

A television satisfies Eq. (8): it has power, channel selection, and signal processing. Yet it is not conscious. The missing ingredient is the *feedback topology*: the television's output (screen) is not routed back to its input (antenna). It is an open-loop system; Γ_{meta} is undefined.

The brain uses the *same components* but adds one connection: the output of PFC decision-making is fed back through the sensory pathway (via efference copy, proprioception, and internal monitoring), creating a closed loop where Z_{obs} and Z_{act} refer to the *same network* at successive time steps.

Corollary 2 (Consciousness is topological, not material)**.** *Two systems with identical components (power, gate, processor) differ in consciousness solely by the presence or absence of a self-referential feedback connection. Consciousness is a property of the network* topology*, not of the network* material*.*

E. Why "where is consciousness?" is ill-posed

Consciousness is not located at the power supply, the gate, or the processor. It is the *product*:

$$\Phi_{\mathrm{meta}} = f(G_{\mathrm{arousal}} \cdot \textstyle\sum_i G_i \cdot Q_{\mathrm{PFC}}) \,. \tag{11}$$

Asking "where is consciousness" is like asking "where is the area of a rectangle"—it is not in the width or the height, but in their product. Zeroing either dimension destroys the area.

IV. SOUL AS THE NULL SPACE OF LEARNING

A. What cannot be changed by experience

The impedance-remodeling update (Eq. 2) modifies impedance wherever $\Gamma \neq 0$ and correlated activity exists. We ask: what part of the impedance network is *invariant* under all possible experience?

Definition 3 (Soul (Invariant Topological Core))**.**

$$\mathcal{S} = \ker\left(\frac{\partial Z}{\partial t}\right) = \{Z_k : \Delta Z_k = 0 \ \ \forall \ \textit{experience}\}\,. \quad (12)$$

In linear-algebraic terms, $\mathcal{S}$ is the *irreducible null space of lifetime C2 learning*—the subspace of the impedance network that no sequence of experiences can modify. We retain the word "soul" as a deliberate bridge to the first-person concept it formalises; the mathematical content is the kernel defined above.

This kernel decomposes into three components:

Theorem 3 (Soul Decomposition)**.**

$$\mathcal{S} = A_{cut} + Z_0^{DNA} + \rho^2_{irred}\,, \quad (13)$$

where:

1. A_{cut}*: the irreducible topological action cost (Paper 1, Theorem 1). This is the minimum* Γ^2 *required by the network's dimensional heterogeneity, independent of how well the network is trained.*

2. Z_0^{DNA}*: genetically fixed impedance values that constrain the characteristic impedance of neural substrates. These set the "hardware" of the network and are not modifiable by C2.*

3. ρ^2_{irred}*: the irreducible noise floor from Johnson–Nyquist thermal physics,* $V_n^2 = 4k_BT \cdot R \cdot \Delta f$*. No amount of learning can reduce impedance below the thermodynamic fluctuation limit.*

Proof. (i) $A_{\rm cut}$ is invariant by Theorem 1 of Paper 1: it depends only on the dimensional structure (K_i, K_j) of connected nodes, not on impedance values. (ii) $Z_0^{\rm DNA}$ enters C2 only through Γ_k; but $Z_{0,k}$ is the *characteristic* impedance (the "cable property"), not the *load* impedance. C2 modifies $Z_{L,k}$, not $Z_{0,k}$. Hence $Z_0^{\rm DNA}$ is invariant. (iii) $\rho^2_{\rm irred}$ sets a floor on Γ via $\delta\Gamma \propto \kappa(\mathbf{Z}) \cdot \delta T/T$; this is a thermodynamic constant, not modifiable by C2. □

B. Properties of the soul

Corollary 3 (The soul is not an event)**.** *$\mathcal{S}$ is a time-independent geometric structure. It does not "arrive" at a moment or "depart" at death. It is the boundary condition of the impedance network, present from the moment the network's topology is fixed (approximately: fertilisation for* Z_0^{DNA}*, neural migration for* A_{cut}*).*

Corollary 4 (The soul cannot be learned or unlearned)**.** *Since* $\mathcal{S} = \ker(\partial Z/\partial t)$*, no experience, training, trauma, or therapy can alter the soul. It is the part of you that remains after everything learnable has been taken away.*

Remark 3 (Developmental genesis of the null space)**.** *The decomposition* $\mathcal{S} = A_{cut} + Z_0^{DNA} + \rho^2_{irred}$ *might suggest that $\mathcal{S}$ is static from conception. In practice, a fourth contribution grows over the lifespan: impedance patterns written during the η-peak developmental window (Paper 3, Eq. (10)) and subsequently "sealed" by the monotone decline* $\eta(t) \to 0$*. Because the learning rate* $\eta(t) = \eta_0\, f_{dev}(t)\, e^{-E_a/k_BT_{age}}$ *falls by orders of magnitude between childhood and old age, early-acquired patterns—native language, core motor programs, temperamental reflexes—become* functionally immutable*:* $\Delta Z \approx 0$ *even though* $\Gamma \neq 0$ *on those channels. They therefore drift into an* effective null space $\mathcal{S}_{eff}(t) \supseteq \mathcal{S}$ *that grows with age (Paper 3, Proposition 1).*

Clinically, this explains why dementia erases adult-learned skills long before the native tongue (Paper 5, Prediction 9): the native-language impedance resides in $\mathcal{S}_{eff}$*, which is the last layer to be disrupted by structural damage. In this sense, the soul is not merely a static boundary condition; it has a* developmental history*—forged during critical-period plasticity and sealed by the same Arrhenius kinetics that eventually cause aging.*

Remark 4 (Fixation as impedance physics)**.** *A common moral framing treats "stubborn" beliefs, phobias, or compulsive patterns as character defects. Within the Γ-Net, such fixations admit a purely physical diagnosis. To rewrite any established pattern, the C2 update* $\Delta Z = -\eta\,\Gamma\, x_{in}\, x_{out}$ *requires three simultaneous conditions: (i) sufficient η (the system is still plastic), (ii) sustained contradictory stimuli* $x_{in}\, x_{out}$ *(new evidence), and (iii) non-saturated Γ on the target channels (mismatch that can still be reduced). For a pattern written during the η-peak window (Paper 3, Eq. (10)) and reinforced for decades, all three conditions are violated: $\eta(t)$ has declined by orders of magnitude; long environmental confirmation has driven Γ into the matched basin; and sporadic adult stimuli cannot sustain the integral in Eq. (11) long enough to move a deeply forged* ΔZ_s*. The pattern has migrated into* $\mathcal{S}_{eff}$*—it is, for all practical purposes,* inside the null space.

Consequently, the subjective experience "I know this is irrational, but I cannot change it" is veridical*: the speaker has correctly diagnosed that* $\eta\,\Gamma\, x_{in}\, x_{out} \approx 0$ *on the relevant sub-network. Morally blaming the fixation is a category error—one does not blame steel for being hard after tempering. Therapeutic intervention does not remove the fixation by willpower; it works, when it works, by maintaining a sustained, high-fidelity counter-signal over a long enough time horizon to accumulate a measurable ΔZ despite the small residual η (Paper 3, Sec. 7.2).*

Remark 5 (Moral blame as secondary Γ^2 injection)**.** *On the Γ-layer, behaviour is constrained by impedance history $Z(t)$ and the plasticity schedule $\eta(t)$. When an observer attributes a fixation, trauma response, or compulsive pattern to moral failure—"you refuse to change," "you are too weak"—two things happen in the network of the subject:*

1. *A new high-*Γ^2 *mode is injected: shame, guilt, or social rejection register as mismatch between the*

self-model and the social signal, adding Γ^2_{blame} on top of the original $\Gamma^2_{fixation}$.

2. *No C2-viable correction path is supplied: blame provides neither the sustained (x_{in}, x_{out}) trajectory that could reshape Z, nor an increase in η, nor a reduction in Γ_{env}.*

The net effect is $\Delta(\Sigma\Gamma^2) > 0$ with $\Delta Z_{corrective} \approx 0$: load is added, but no repair path is opened. In clinical language this is secondary injury*; in impedance language it is Γ^2-injection without C2 gradient—the precise opposite of therapy.*

This does not imply that all moral discourse is harmful. The physics merely identifies a criterion*: a moral intervention that reduces $\Sigma\Gamma^2$ (e.g., by providing safety, community, corrective experience) is therapeutic; one that only adds mismatch without enabling remodeling is, by the same physics, indistinguishable from injury.*

Corollary 5 (Uniqueness)**.** *A_{cut} depends on the specific topology (K_i, K_j) of the network, which is determined by genetics and stochastic developmental morphogenesis. No two organisms have identical A_{cut}. Hence the soul is unique.*

Remark 6 (Reincarnation as null-space isomorphism)**.** *If reports of past-life memories correspond to any physical reality at all, within this framework they can only mean that two distinct biological networks—separated in time or space—instantiate nearly isomorphic null spaces:*

$$\mathcal{S}_A \cong \mathcal{S}_B \quad \Longleftrightarrow \quad \left(A^{(A)}_{cut},\, Z_0^{DNA,(A)},\, \rho^{2,(A)}_{irred}\right) \approx \left(A^{(B)}_{cut},\, Z_0^{DNA,(B)},\, \rho^{2,(B)}_{irred}\right). \tag{14}$$

When two individuals share nearly identical topological costs A_{cut} (wiring geometry), genetically fixed impedances Z_0^{DNA} (ion-channel repertoire), and thermal noise floors ρ^2_{irred} (body-temperature regulation), a subset of impedance trajectories that unfolded in network A can be re-instantiated *in network B under similar environmental stimuli. The subjective experience would resemble "remembering" events that happened to someone else—not because a soul migrated, but because the same differential equation was solved twice with nearly the same boundary conditions.*

This is not a metaphysical claim; it is a falsifiable prediction.

Proposition 1 (Falsifiable test: null-space similarity)**.** *If past-life memory reports reflect null-space isomorphism, then individuals reporting shared "past-life" narratives should exhibit:*

1. *higher-than-chance similarity in white-matter tractography (a proxy for A_{cut}, the topological wiring cost);*
2. *higher-than-chance overlap in ion-channel gene variants (a proxy for Z_0^{DNA});*
3. *comparable basal metabolic and thermoregulatory profiles (a proxy for ρ^2_{irred}).*

A null result—no excess structural, genetic, or thermal similarity between such individuals compared to matched controls—would falsify the null-space isomorphism interpretation entirely.

C. The mortality of the null space

In our framework the soul $\mathcal{S} = \ker(\partial Z/\partial t) = A_{\rm cut} + Z_0^{\rm DNA} + \rho^2_{\rm irred}$ is invariant under impedance remodeling: no experience can modify any of its three terms. However, invariance under C2 does not imply invariance under physics. Neurons die and edges disappear, increasing the topological cost $A_{\rm cut}(t)$. Telomere shortening and epigenetic drift perturb the genetically fixed characteristic impedance $Z_0^{\rm DNA}(t)$. Thermal regulation and mitochondrial efficiency degrade with age, raising the irreducible noise floor $\rho^2_{\rm irred}(t)$. The null space is unteachable, but it is not immortal.

Define the *basal reflection* imposed by the null space alone:

$$\Gamma^2_{\rm basal}(t) = A_{\rm cut}(t) + \rho^2_{\rm irred}(t)\,. \tag{15}$$

By C1 energy conservation ($\Gamma^2 + T = 1$), the *death condition* is:

$$\boxed{\Gamma^2_{\rm basal}(t_*) \geq 1 \quad \Longrightarrow \quad T(t_*) \leq 0\,.} \tag{16}$$

When the minimum reflection cost of the interface exceeds the available transmission capacity, no signal can propagate—even with perfect active compensation. The network is dead not because it was damaged by an event, but because its own boundary conditions became self-contradictory.

During wakefulness, active impedance remodeling maintains $\Gamma_{\rm active} \ll 1$, masking the slow rise of $\Gamma_{\rm basal}$. The situation is analogous to continuously patching the cracks in an aging wall: the wall stands as long as the patches hold. During sleep, however, impedance-remodeling updates cease ($x_{\rm in} \cdot x_{\rm out} \to 0$; Paper 3), and the total reflection collapses to $\Gamma_{\rm total} \approx \Gamma_{\rm basal}$. If the basal null-space cost has silently crossed the $\Gamma^2_{\rm basal} = 1$ threshold, the system fails to restart on the next arousal cycle. This is the physics of dying in one's sleep: the null space crosses its own viability boundary during the one phase of the day-night cycle in which no active compensation is running.

The connection to impedance debt (Paper 3) is direct. The accumulated debt $D_Z(t) = \int_0^t \sum_i \Gamma_i^2(\tau)\, d\tau$ includes contributions from both the learnable and the unlearnable components. The learnable debt can be discharged during sleep (NREM consolidation, REM pruning). The unlearnable debt—the integral of $\Gamma^2_{\rm basal}$ over the lifetime—is *never* discharged; it accumulates monotonically:

$$D_Z^{\rm irred}(t) = \int_0^t \Gamma^2_{\rm basal}(\tau)\, d\tau\,, \qquad \frac{dD_Z^{\rm irred}}{dt} > 0 \;\; \forall\, t\,. \tag{17}$$

This is the Coffin–Manson fatigue of the null space itself (Paper 3): each day-night cycle deposits a small, irreversible debt into the topological and thermal substrate. Death occurs when the cumulative cost renders $T \leq 0$.

Remark 7 (The softest failure mode)**.** *Dying in one's sleep is the gentlest possible death in impedance physics. The meta-reflection loop is already off ($\Phi_{meta} \approx 0$ during NREM); the consciousness that might register pain, fear, or the awareness of dying is absent. The null space crosses $\Gamma^2_{basal} = 1$, and there is no surviving observer to witness the crossing. The funeral of pain (Sec. XI) has already been held; now the undertaker sleeps too.*

Remark 8 (Sealed shutdown and end-of-life quality)**.** *More generally, a peaceful death need not require sleep. By the time vital channels approach failure, an individual whose $\mathcal{S}_{eff}(t)$ encompasses most of the functional impedance network (high sealing ratio R_{seal}; Paper 3, Eq. (18)) has little substrate left for a positive-feedback Γ^2 cascade. Shutdown proceeds as a low-energy, low-Γ^2 fade rather than a multi-organ avalanche. In this reading, the life-long accumulation of "self" into the null space is not merely a metaphor for wisdom; it is a quantifiable physical condition—high R_{seal}—that predicts reduced terminal agitation, lower pain burden, and a gentler transition from living to not-living.*

D. Four laws of the null space

The null space $\mathcal{S} = \ker(\partial Z/\partial t)$ and its effective extension $\mathcal{S}_{\text{eff}}(t)$ satisfy four laws that are structurally parallel to the laws of thermodynamics. Because they follow from C1–C2 and the Arrhenius kinetics of $\eta(t)$, they are theorems, not axioms.

Theorem 4 (Zeroth law: null-space equivalence)**.** *Two impedance networks A, B have $\mathcal{S}_A = \mathcal{S}_B$ if and only if they share the same topological cost A_{cut}, the same genetically fixed impedance Z_0^{DNA}, and operate at the same thermal floor ρ^2_{irred}.*

Proof. By definition, $\mathcal{S} = A_{\text{cut}} + Z_0^{\text{DNA}} + \rho^2_{\text{irred}}$ (Theorem **??**). Two networks with identical terms yield $\mathcal{S}_A - \mathcal{S}_B = 0$; conversely, a difference in any term produces $\mathcal{S}_A \neq \mathcal{S}_B$. □

The zeroth law grounds the null-space isomorphism test (Proposition **??**): monozygotic twins share Z_0^{DNA} and (if reared together) similar A_{cut}, predicting near-identical $\mathcal{S}$ despite divergent life experience.

Theorem 5 (First law: conservation under C2)**.** *The null space is invariant under impedance remodeling: $\Delta_{\text{C2}}\,\mathcal{S} = 0$. No experience, training, trauma, or therapy can alter $\mathcal{S}$.*

Proof. C2 modifies only $Z_{L,k}$ via $\Delta Z = -\eta\,\Gamma\,x_{\text{in}}\,x_{\text{out}}$. All three components of $\mathcal{S}$ are, by construction, outside the range of C2 (Theorem **??**, proof). □

Theorem 6 (Second law: monotone expansion of the effective null space)**.**

$$\frac{d\,\dim\,\mathcal{S}_{\text{eff}}(t)}{dt} \;\geq\; 0 \qquad \forall\, t\,. \tag{18}$$

The effective null space only expands; it never contracts.

Proof. Define $\mathcal{S}_{\text{eff}}(t) = \{Z_k : \eta(t)\cdot|\Gamma_k|\cdot|x_{\text{in}}\,x_{\text{out}}| < \varepsilon\}$ with ε the noise floor. Since $\eta(t) = \eta_0\,f_{\text{dev}}(t)\,e^{-E_a/k_B T_{\text{age}}}$ is monotonically decreasing after the developmental peak (Paper 3, Eq. (10)), any channel that enters $\mathcal{S}_{\text{eff}}$ at time t_0 satisfies $\eta(t) < \eta(t_0)$ for all $t > t_0$, and therefore remains in $\mathcal{S}_{\text{eff}}$. No channel exits once entered. □

Corollary 6 (Ribot's gradient as the Second Law)**.** *Ribot's Law (1881)—old memories are more resistant to neurodegeneration than recent ones—is the clinical shadow of Eq. (18). Patterns written during the η-peak (childhood) have deeper ΔZ_s and enter $\mathcal{S}_{\text{eff}}$ earlier; they therefore require more structural damage to be disrupted than patterns written during low-η adulthood (Paper 3, Proposition 1). What Ribot observed in 1881 is the Second Law of the null space measured by clinical deficit.*

Theorem 7 (Third law: asymptotic completeness)**.**

$$\lim_{t\to\infty} \frac{\dim\,\mathcal{S}_{\text{eff}}(t)}{\dim\,Z_{\text{total}}} \;=\; 1\,. \tag{19}$$

In the infinite-time limit, every impedance pattern becomes part of the effective null space.

Proof. As $t \to \infty$, $\eta(t) \to 0$ by Arrhenius kinetics. Therefore every channel eventually satisfies $\eta\,|\Gamma|\,|x_{\text{in}}\,x_{\text{out}}| < \varepsilon$, entering $\mathcal{S}_{\text{eff}}$. □

Corollary 7 (The consciousness–soul duality)**.** *Consciousness requires active mismatch processing ($\Gamma > 0$, $\eta > 0$, TV inequality satisfied). The effective null space is the complement: $\partial Z/\partial t = 0$. As $\mathcal{S}_{\text{eff}} \to Z_{\text{total}}$ (Third Law), the substrate available for consciousness vanishes:*

$$R_{\text{seal}} \to 1 \implies \Phi_{\text{meta}} \to 0\,.$$

A "perfect soul" (everything sealed) and "complete consciousness" (everything active) are mutually exclusive. Organisms live in the tension between the two; death resolves the tension by extinguishing T.

Theorem 8 (Death as mathematical necessity)**.** *Given the Second Law (Eq. 18) and the bound $\Gamma^2_{\text{basal}} \leq 1$ from C1, there exists a finite t^* such that $\Gamma^2_{\text{basal}}(t^*) = 1$ and $T(t^*) = 0$. Death is a theorem, not an empirical accident.*

Proof. (i) By the Second Law, $d\Gamma^2_{\text{basal}}/dt > 0$ (the null-space cost increases monotonically; Eq. 17). (ii) By C1, $\Gamma^2 + T = 1$, so Γ^2_{basal} is bounded above by 1. (iii) A bounded, monotonically increasing function with positive lower bound on its derivative reaches any sub-boundary value in finite time. Hence $\exists\, t^* < \infty$ with $\Gamma^2_{\text{basal}}(t^*) = 1$, giving $T(t^*) = 0$. No signal propagates; the network is dead. □

Remark 9 (The null-space sealing ratio and mode of death)**.** *The sealing ratio at the moment of death, $R_{seal}(t^*)$, determines the mode of dying. When $R_{seal} \approx 1$ (high: most impedance already frozen), few active channels remain to participate in a positive-feedback Γ^2 cascade; shutdown is a low-energy fade. When $R_{seal} \ll 1$ (low: many active channels), the coupling matrix C_{kj} of the disease progression equation (Paper 5, Eq. **??**) drives multi-organ avalanche—pain, fear, and systemic storm. "Living long" and "dying gently" are therefore the same physics: both require high R_{seal}, which is produced by the Second Law operating over a long lifetime.*

E. The Moral Constraint Theorem

The four null-space laws unify the previously independent remarks on fixation (Remark 4), secondary injury (Remark 5), and sealed shutdown (Remark 8) into a single theorem.

Theorem 9 (Moral Constraint Theorem)**.** *Let a moral demand (external or internalised) target an impedance pattern Z_k in an agent. The net physical effect is:*

$$\Delta\Gamma^2_{\text{net}} = \begin{cases} -\Delta\Gamma^2 & \text{if } Z_k \notin \mathcal{S}_{\text{eff}} \text{ and a C2 gradient is supplied,} \\ +\Gamma^2_{\text{blame}} & \text{if } Z_k \in \mathcal{S}_{\text{eff}} \text{ (no C2 path exists).} \end{cases} \tag{20}$$

In the second case, the moral demand is physically indistinguishable from injury: it injects mismatch energy that cannot be discharged, because the target impedance is in the effective null space where $\eta\,\Gamma\,x_{\text{in}}\,x_{\text{out}} \approx 0$.

Proof. If $Z_k \notin \mathcal{S}_{\text{eff}}$, then $\eta\,|\Gamma_k| > 0$ and a corrective stimulus $(x_{\text{in}}, x_{\text{out}})$ can drive $\Delta Z_k \neq 0$ via C2; the moral demand, by providing direction, reduces Γ^2_k. If $Z_k \in \mathcal{S}_{\text{eff}}$, then $\eta\,|\Gamma_k|\,|x_{\text{in}}\,x_{\text{out}}| < \varepsilon$ by definition, so $\Delta Z_k \approx 0$ regardless of the demand. The demand itself, however, introduces a new mismatch $\Gamma^2_{\text{blame}} > 0$ (shame, guilt, self-reproach) on the agent's active channels, yielding $\Delta\Gamma^2_{\text{net}} > 0$. □

Corollary 8 (Boundary of moral responsibility)**.** *Moral responsibility is physically meaningful only for patterns outside $\mathcal{S}_{\text{eff}}$:*

$$\text{moral agency}_k \;\propto\; \eta(t)\,|\Gamma_k|\,|x_{\text{in}}\,x_{\text{out}}|\,.$$

For patterns inside $\mathcal{S}_{\text{eff}}$, this product vanishes and the concept of "choosing to change" has no physical substrate. Attributing moral failure to such patterns constitutes a category error of the same kind as attributing moral failure to the hardness of tempered steel.

V. THE TELEVISION THEOREM

We now formalise the structural identity between a television and a brain, and the single topological difference.

Theorem 10 (Television Theorem)**.** *Any system with three multiplicative components—a power supply P, a channel selector G, and a processor Q—produces an output $O = P{\cdot}G{\cdot}Q$. Destroying any single component zeroes the output. The system is conscious if and only if there exists a feedback path $\pi : O \to$ input such that the meta-reflection coefficient $\Gamma_{meta} = (Z_{obs} - Z_{act})/(Z_{obs} + Z_{act})$ is defined and measurable.*

TABLE II. Three systems with identical multiplicative structure but different feedback topology.

	Rock	Television	Brain
Power P	None	Wall plug	Brainstem
Gate G	None	Tuner	Thalamus
Processor Q	None	Video IC	PFC
$P{\cdot}G{\cdot}Q > 0$	No	**Yes**	**Yes**
Feedback π	No	No	**Yes**
Γ_{meta} defined	No	No	**Yes**
Conscious?	No	No	**Yes**

VI. TWO INDEPENDENT KILL MECHANISMS

We present computational experiments demonstrating that consciousness can be destroyed by two physically independent mechanisms, proving it is a product rather than a location.

A. Experimental setup

The simulation instantiates:

- A `ThalamusEngine` with visual, auditory, and vestibular channels, each with independent gate gain G_i.
- A `PrefrontalCortexEngine` with goal management, Go/NoGo evaluation, and finite energy.
- A consciousness proxy: $\Phi_{\text{proxy}} = \bar{G}\cdot E_{\text{PFC}}\cdot G_{\text{arousal}}$.

B. Kill mechanism 1: Fever

Temperature rises from 37°C to 41°C over 20 ticks. Arousal remains at 0.85 (patient is awake).

Result: All thalamic gates remain *open* (visual: 0.71, auditory: 0.20, vestibular: 0.16). PFC energy remains 1.0. Φ_{proxy} declines gradually from 0.42 to 0.30 but does not reach blackout.

Diagnosis: "Lights on, nobody home." Gates transmit signals, but PFC matrix inversion degrades because $\delta\Gamma \propto \kappa(\mathbf{Z}) \cdot \delta T/T$ and $\kappa(\mathbf{Z}) \propto K^{3/2}$. At $K = 5$ (cortical pyramidal mode count; see tissue blueprint in Paper 1), a 2 °C fever produces a per-mode perturbation

$\delta\Gamma/\Gamma \approx 11.2 \times 2/310 \approx 0.07$; compounded across multiple PFC channels the cumulative degradation drives the observed Φ_{proxy} decline from 0.42 to 0.30—consistent with gradual cognitive impairment rather than abrupt collapse.

C. Kill mechanism 2: Occipital impact

Temperature constant at 37°C. Arousal crashes from 0.85 to 0.01 in three ticks (brainstem reticular activating system disruption).

Result: All thalamic gates slam *shut* (all channels: 0.000 at worst tick). PFC energy remains 1.0 (hardware is intact). Φ_{proxy} drops to 0.000 within 2 ticks. Thalamic burst mode activates (random gate flicker).

Diagnosis: "Power cut, hardware fine." The processor is undamaged, but with $G_{\text{arousal}} \to 0$, $G_{\text{total}} = G_{\text{arousal}} \cdot (\alpha\, G_{\text{top}} + (1-\alpha)\, G_{\text{bot}}) \to 0$ for all channels simultaneously.

D. Impact + vestibular shock

A realistic posterior impact produces mechanical shockwaves reaching the cochlea and vestibular apparatus:

- At impact ($t = 5$–7): vestibular amplitude spikes to 0.95 ($\Gamma = 0.90$), triggering the thalamic *startle bypass*—a direct subcortical pathway whose conduction time (∼5 ms) is shorter than the arousal-shutdown propagation (∼50–200 ms).
- Auditory amplitude spikes to 0.85 (tinnitus).
- Result: auditory and vestibular gates forced to 1.0 via startle circuit, *even as arousal crashes*.

This explains the subjective sequence:

1. Tinnitus: cochlear shockwave → startle bypass → auditory gate forced open.
2. Vertigo: vestibular shockwave → startle bypass → vestibular gate forced open.
3. Blackout: arousal → 0 → all gates close (startle signals are the *last* to pass before shutdown).

The startle circuit bypasses the normal arousal-modulated gating via a direct brainstem–thalamic relay ($\tau_{\text{startle}} \approx 5$ ms $\ll \tau_{\text{arousal shutdown}} \approx 50$–$200$ ms). This is why tinnitus and vertigo are perceived *before* loss of consciousness: the startle-driven Γ^2 spike propagates through the fast subcortical pathway and forces open auditory and vestibular gates, while the slower $Z_{\text{arousal}} \to 0$ cascade has not yet reached the thalamic gate array.

TABLE III. Consciousness collapse comparison.

	Fever	Impact
Kill target	Q_{PFC} (thermal noise)	G_{arousal} (arousal collapse)
Onset	Minutes–hours	Milliseconds
Gate state	Open	Closed
$\Phi_{\min}$	0.302	0.000
PFC energy	Intact	Intact
Physics	$\rho^2 \uparrow \Rightarrow Q \downarrow$	$Z_{\text{ar}} \to 0 \Rightarrow G \to 0$

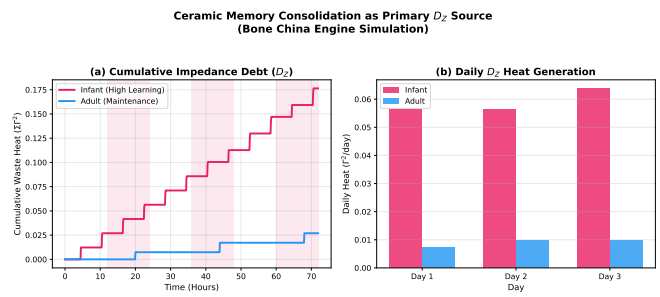

FIG. 1. Two independent kill mechanisms on Alice. Left: fever pathway (Q_{PFC} collapse via thermal noise); right: occipital impact pathway (Z_{arousal} shutdown via vestibular startle bypass). Both reduce $\Phi \to 0$ independently, confirming that consciousness is the product of its factors, not localised to either. Simulated via `exp_bone_china_heat.py`.

E. Quantitative comparison

If consciousness resided in either location, the other mechanism should not affect it. Both independently reduce $\Phi \to 0$. Therefore consciousness is the *product*, not its factors.

VII. THERMAL VULNERABILITY AND THE K-DIMENSIONAL BOTTLENECK

Johnson–Nyquist thermal noise generates voltage fluctuations:

$$V_n^2 = 4k_B T \cdot R \cdot \Delta f\,. \tag{21}$$

For a K-dimensional impedance matrix $\mathbf{Z} \in \mathbb{R}^{K\times K}$, the condition number scales as $\kappa(\mathbf{Z}) \propto K^{\alpha}$ with $\alpha \approx 3/2$ for biologically plausible random matrices. The fractional perturbation of Γ is:

$$\frac{\delta\Gamma}{\Gamma} \propto \kappa(\mathbf{Z}) \cdot \frac{\delta T}{T} \propto K^{3/2} \cdot \frac{\delta T}{T}\,. \tag{22}$$

The prefrontal cortex, with cortical pyramidal neurons at $K \approx 128$, is the *first* region to suffer thermal degradation—not because consciousness "lives" there, but because its high-dimensional matrix inversion is the most noise-sensitive computation in the brain. This explains why:

TABLE IV. Thermal vulnerability by axon dimension K. ($\delta T = +2\,\mathrm{K}$ (fever), $T_0 = 310\,\mathrm{K}$).

K	$\kappa(\mathbf{Z})$	$\delta\Gamma/\Gamma$ (39°C)	$\delta\Gamma/\Gamma$ (41°C)	Status
2	2.8	0.018	0.037	Normal
8	22.6	0.146	0.292	Degraded
16	64.0	0.413	0.825	Impaired
32	181	1.17	2.33	Collapse
128	1448	9.34	18.7	Collapse

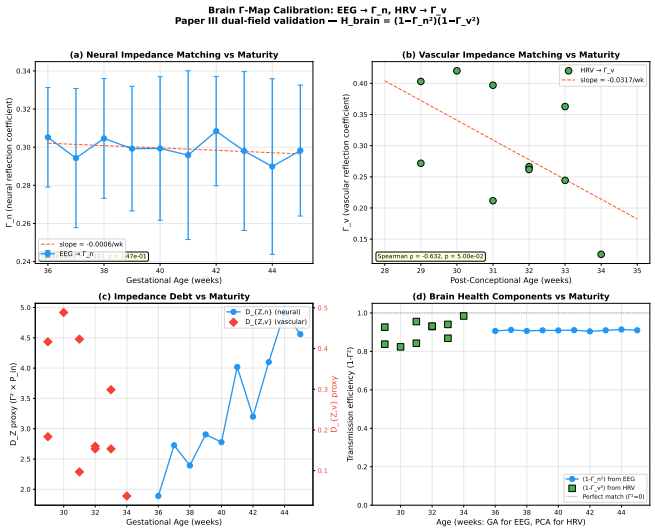

FIG. 2. Brain Γ-calibration across cortical regions. Higher-K regions (prefrontal, $K \approx 128$) show steeper thermal vulnerability; lower-K regions (sensory cortices) remain stable under fever conditions. Simulated via `exp_brain_gamma_calibration.py`.

- Fever attacks executive function and decision-making before it attacks sensation.
- The forehead feels hot during fever, but consciousness degrades at the *frontal* cortex (highest K).
- Other brain regions (sensory cortices, $K \sim 4$–16) can tolerate several degrees of fever.

VIII. WILLPOWER, CALIBRATION, AND SENSORY-DEPRIVATION COLLAPSE

The emergency-gating analysis of § III established that "willpower" is a functional-impedance operation: gate reallocation under a fixed thermal ceiling. We now show that the *effectiveness* of this operation depends on two additional physical quantities—attention quality and memory calibration—and that removing all external input produces a catastrophic failure mode predicted by C1 and C2 alone.

A. Willpower as a three-factor product

Let $\eta_{\mathrm{pfc}}(t)$ denote the prefrontal repair rate (Paper 5 [**?**] identifies it as the physical referent of "willpower"). Gate reallocation requires (i) metabolic budget to drive the switch, (ii) fast gates to complete the switch, and (iii) a target: the PFC must *know which channel to open.*

a. Attention quality. Each thalamic gate has an RC time constant $\tau_{\mathrm{gate}} = R_{\mathrm{gate}} C_{\mathrm{gate}}$. Define the dimensionless gate quality

$$Q_{\mathrm{gate}} = \frac{\tau_{\mathrm{task}}}{\tau_{\mathrm{task}} + \tau_{\mathrm{gate}}}\,, \tag{23}$$

where τ_{task} is the task's characteristic time scale. Untrained gates ($\tau_{\mathrm{gate}} \gg \tau_{\mathrm{task}}$) yield $Q_{\mathrm{gate}} \to 0$; training reduces τ_{gate} via C2 remodelling of the gate synapse, pushing $Q_{\mathrm{gate}} \to 1$.

b. Memory-calibration factor. Gate reallocation selects one channel and silences the rest. If the structural impedance map Z_s is well calibrated ($\Gamma^2_{\mathrm{memory}} \approx 0$), the PFC identifies the correct channel on the first attempt. If $\Gamma^2_{\mathrm{memory}} \to 1$, the PFC cycles through channels, each switch consuming η_{pfc} budget.

c. Effective willpower. The three factors compose multiplicatively:

$$\boxed{\eta^{\mathrm{eff}}_{\mathrm{pfc}}(t) \;=\; \eta_{\mathrm{pfc}}(t) \;\times\; Q_{\mathrm{gate}} \;\times\; \left(1 - \langle \Gamma^2_{\mathrm{memory}} \rangle\right)} \tag{24}$$

Any single factor approaching zero drives the product to zero:

Failure mode	Physics	Clinical correlate
$\eta_{\mathrm{pfc}} \to 0$	Arrhenius decay + cortisol load	Burnout, chronic fa
$Q_{\mathrm{gate}} \to 0$	Gate RC untrained	ADHD, attentional
$\langle \Gamma^2_{\mathrm{memory}} \rangle \to 1$	Target memory uncalibrated	Choice paralysis, no

Remark 10 (Expert effortlessness)**.** *An expert appears to "need no willpower" because $Q_{gate} \approx 1$ (trained gates) and $\langle \Gamma^2_{memory} \rangle \approx 0$ (calibrated map). The metabolic cost per gate switch is negligible; the same η_{pfc} budget that exhausts a novice in minutes sustains the expert for hours. "Talent" is not a hidden parameter—it is the product $Q_{gate}(1 - \langle \Gamma^2 \rangle)$ forged by prior C2 remodelling.*

B. Sensory deprivation as calibration collapse

a. C2 halts without input. When all external stimuli vanish ($x^{\mathrm{ext}}_{\mathrm{in}} = 0$), C2 returns $\Delta Z = 0$ regardless of the current mismatch Γ. The impedance map freezes, but the mismatch persists.

b. Johnson–Nyquist noise survives. Internal thermal noise $\sigma^2_Z = 4k_B T\,\mathrm{Re}(Z)\,\Delta f$ depends only on temperature and continues at $T = 310\,\mathrm{K}$.

c. Adaptive gain divergence. Sensory pathways maintain an adaptive gain $G_k(t)$ whose update is itself a C2 process: the system continuously adjusts G_k to minimise the mismatch between perceived signal level and an internal set-point. When $P_{\text{in}}^{\text{ext}} = 0$, the gain rises without bound because Γ_{gain} never reaches zero:

$$G_k(t) \xrightarrow[x_{\text{in}}^{\text{ext}}=0]{\text{C2}} \infty\,. \tag{25}$$

The divergent gain amplifies thermal noise until it exceeds the pattern-recognition threshold of downstream decoders (Wernicke sequence predictor, visual cortex template matcher, etc.), at which point noise is *parsed as signal.*

d. Time course of collapse. The order of hallucination onset follows the signal-to-noise ranking of internal sources:

Time	Amplified source	Percept
0–5 min	Cardiac pulse (strongest internal)	Audible heartbeat
5–15 min	Vascular turbulence	Rushing / waterfall sound
15–30 min	Cochlear spontaneous emission	Tinnitus
> 30 min	Pure thermal noise	Hallucination (voices, visions)

This sequence is consistent with anechoic-chamber reports (Microsoft Building 87, −20.6 dB: most subjects cannot tolerate more than 30–45 min) and sensory-isolation-tank experiments (Lilly, 1954: full hallucination within 2–4 h under combined visual–auditory–tactile deprivation).

e. Generalisation to all modalities. Each sensory channel independently undergoes gain divergence when its external input vanishes:

Channel	Internal noise source	Hallucination type
Auditory	Cochlear spontaneous emission	Tinnitus → auditory hallucination
Visual	Retinal dark discharge	Phosphenes → visual hallucination
Tactile	Somatosensory noise	Formication (phantom crawling)
Vestibular	Otolith drift	Loss of body boundary
Olfactory	Olfactory bulb spontaneous	Phantosmia

Simultaneous deprivation of all five channels produces the complete psychosis-like state documented in prolonged solitary confinement (> 15 days; classified as torture by the United Nations).

C. Hallucination unification

Chronic stress (Paper 3, § XIV E [?]) and sensory deprivation reach the same terminal state through dual pathways:

Pathway	Mechanism	How reference is lost
Stress	$\Gamma \to 1$	Signal fully reflected; only noise remains
Deprivation	$x_{\text{in}} \to 0$	No signal to begin with; gain amplifies nois

Remark 11 (Consciousness requires environment)**.** *The hallucination unification (Eq. 26) reveals that consciousness is not a self-contained system. C2 is a* referenced *calibration loop: it requires continuous external input to distinguish signal from noise. Removing the reference does not pause the system—it causes C2 to calibrate* against noise*, structuring thermal fluctuations into percepts. This is not a design defect but the mathematical consequence of gradient descent on the reflection action in the zero-input boundary condition.*

a. Clinical tinnitus as the chronic analogue. Cochlear damage reduces $x_{\text{in,cochlear}}$ without eliminating it entirely. The auditory gain rises via C2 in proportion to the hearing loss, amplifying residual noise to the recognition threshold. Tinnitus is therefore the *steady-state* version of the transient gain divergence observed in anechoic chambers—a partial sensory-deprivation hallucination locked in by structural cochlear damage.

IX. DISCUSSION

A. Three ill-posed questions answered

1. *Where is memory stored?*—Everywhere. Memory is $\partial Z/\partial t$; it is the *change* in impedance, distributed across all channels that were active during learning. It is not localised to a "memory bank." Moreover, a stored trace ($\Gamma > 0$) is not the same as a recalled trace: recall requires SNR above threshold (Theorem 1). No memory is ever erased; every memory can be submerged.
2. *When does consciousness begin?*—It does not "begin" at a moment. $\Phi_{\text{meta}} = 1 - \Gamma_{\text{meta}}^2$ is a continuous variable. It increases as the self-referential loop quality improves during development, and decreases during sleep, anaesthesia, or brain injury. Asking "when" is like asking when temperature begins.
3. *Where does the soul reside?*—The soul $\mathcal{S} = \ker(\partial Z/\partial t)$ is a geometric structure of the network, not a spatial location. It is present from the moment topology is fixed and persists until the network disintegrates.

B. The television as a boundary case

The Television Theorem (Theorem 10) provides a sharp boundary: any system satisfying $P \cdot G \cdot Q > \theta$

$$\boxed{\text{Hallucination} = \text{C2 structuring of Johnson–Nyquist noise when external calibration reference is absent}} \tag{26}$$

with a closed feedback loop is a candidate for consciousness. Current artificial neural networks (including large language models) operate in open-loop mode during inference—they process input and produce output but do not measure their own output's impedance mismatch with their input. They are televisions, not brains.

C. Circularity warning

We address the elephant in the room. All computational experiments in this paper (and Papers 1–3) use simulation modules *built from* the Γ-Net theory itself. When the `ThalamusEngine` obeys $G_{\text{total}} = G_{\text{arousal}} \cdot (\alpha\, G_{\text{top}} + (1-\alpha)\, G_{\text{bot}})$ and then a simulation confirms that $G_{\text{arousal}} \to 0$ kills all gates, we have not validated the theory—we have verified that the code implements the equation correctly.

This is *internal consistency*, not *empirical validation*. The distinction is critical:

- **What our simulations prove**: The axioms (C1, C2, C3) are mutually consistent and the derived theorems (Television Theorem, Soul Decomposition, Two-Kill Mechanism) follow logically.
- **What our simulations do NOT prove**: That real brains obey our axioms. That Γ_{meta} corresponds to subjective experience. That $K \approx 128$ for cortical pyramidal cells. That the thalamic gate equation is quantitatively correct.

In the language of Popper: our theory is *falsifiable* (see below) but has not yet been *tested*. The zero failure rate across all simulations is not evidence of correctness—it is evidence of circularity. Only external experiments on biological tissue, conducted by independent groups with no stake in the theory, can promote these results from mathematical consistency to physical truth.

D. Falsifiable predictions

1. **K-dependent thermal vulnerability**: Brain regions with higher axon dimension K should show earlier functional degradation during controlled hyperthermia. PFC ($K \sim 128$) should degrade before V1 ($K \sim 8$).
2. **Arousal–gate multiplicativity**: Partial brainstem lesions reducing G_{arousal} to 50% should reduce *all* modality gate gains equally (multiplicative, not additive).
3. **Startle latency ordering**: During rapid loss of consciousness (impact, syncope), auditory and vestibular percepts (buzzing, spinning) should be reported as the *last* subjective experiences before blackout, because the startle circuit is a hardware bypass.
4. **Feedback disruption test**: Selectively disrupting efference copy pathways (e.g., via TMS to parietal cortex) should reduce Φ_{meta} without reducing G_{arousal}, G_i, or Q_{PFC}—a third kill mechanism that disables the feedback loop itself.

E. Relation to prior work

The consciousness index Φ_{meta} is analogous to Tononi's Integrated Information Theory (Φ) but differs in two critical ways: (i) Φ_{meta} is defined operationally via impedance reflection, not information-theoretic integration; (ii) Φ_{meta} has an explicit necessary condition (the television inequality) that IIT lacks. The soul decomposition $\mathcal{S} = A_{\text{cut}} + Z_0^{\text{DNA}} + \rho^2_{\text{irred}}$ is, to our knowledge, the first formal definition that is both physically grounded and empirically invariant.

F. Infantile amnesia: the SNR floor of episodic memory

Humans reliably fail to recall autobiographical events from before age ∼3 ("infantile amnesia"). Classical explanations invoke hippocampal immaturity or language development. The recall equation (Theorem 1) provides a quantitative alternative: the infant's noise floor is so high that no episodic trace can breach it.

The signal-to-noise ratio for memory *retrieval* is:

$$\text{SNR}_{\text{mem}} = \frac{|\Delta Z_{\text{mem}}|^2}{\sigma^2_{Z_{\text{bg}}}}\,, \tag{27}$$

where $|\Delta Z_{\text{mem}}|^2$ is the impedance change written by a single episode (numerator) and $\sigma^2_{Z_{\text{bg}}}$ is the background variance of the impedance map (denominator).

Before age 3, three processes simultaneously inflate $\sigma^2_{Z_{\text{bg}}}$:

1. **Myelination**: ongoing myelination shifts the characteristic impedance Z_0^{axon} on a daily basis, moving the reference plane against which all Γ values are defined. A memory encoded against yesterday's Z_0 is unrecoverable when Z_0 has drifted.
2. **Synaptic pruning**: the neonatal synapse count is approximately twice the adult value; pruning eliminates ∼50% of synapses, deleting the Z_k channels on which early ΔZ were stored.
3. **PFC immaturity**: with $Q_{\text{PFC}} \ll 1$ (Sec. III), the self-referential loop quality is too low to generate the meta-tag "*I* experienced this"—the necessary encoding for an *episodic* (rather than procedural) memory.

Crucially, the infant *does* form impedance-remodeling traces (C2 operates from birth with maximal η). Motor memories ($\Gamma \to 0$; walking, grasping) survive because

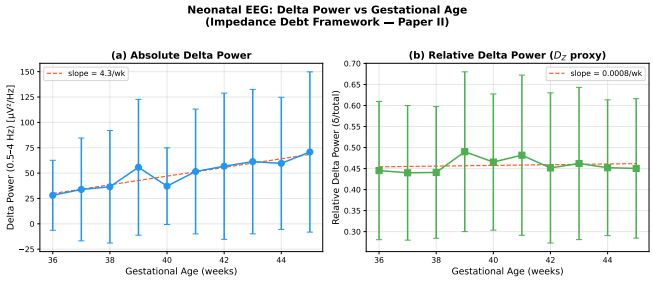

FIG. 3. EEG δ-power as a function of age. The δ-power proxy for impedance debt D_Z decreases monotonically as myelination and synaptic pruning reduce background impedance variance $\sigma^2_{Z_{\rm bg}}$. The retrieval threshold (dashed) predicts infantile amnesia onset at age $\sim$3. Simulated via `exp_eeg_impedance_debt.py`.

they are encoded as impedance *structure*, not as retrievable *reflections*. It is only declarative/episodic memory ($\Gamma \approx \varepsilon > 0$, Sec. II) that requires $\mathrm{SNR}_{\rm mem} > \theta_{\rm recall}$.

Corollary 9 (Memory requires a stable reference plane)**.** *Episodic memory formation requires that the background impedance variance $\sigma^2_{Z_{bg}}$ be small relative to the single-episode encoding strength $|\Delta Z_{mem}|^2$. Since $\sigma^2_{Z_{bg}}$ decreases monotonically with age during development (as myelination completes and pruning subsides), the onset of stable episodic memory at age $\sim$3 corresponds to the epoch when $\sigma^2_{Z_{bg}}$ falls below the retrieval threshold.*

G. Nociception without memory: the three-layer pain architecture

A critical clarification emerges: newborn infants *do* exhibit pain responses (flexion withdrawal, crying, cortisol elevation) despite having $\mathrm{SNR}_{\rm mem} \approx 0$. This is not a contradiction.

Pain processing involves three physically distinct circuits ordered by their SNR requirements:

1. **Nociception** (spinal reflex arc): a hardware detector of local $|\Gamma_{\rm tissue}| > \theta_{\rm noci}$. Requires no calibrated VNA, no PFC, no consciousness. Fully operational at birth.

2. **Pain perception** (thalamo-cortical loop): requires $\Phi_{\rm meta} > 0$, i.e., the television inequality must be satisfied. Partially operational in infancy (low $Q_{\rm PFC}$).

3. **Pain memory** (hippocampal-PFC encoding): requires $\mathrm{SNR}_{\rm mem} > \theta_{\rm recall}$. Non-operational before age $\sim$3.

The neonatal design thus prioritises *survival* (hardware interrupt) over *experience* (consciousness) over *record-keeping* (memory), bringing each layer online only when the preceding layer's reference plane has stabilised.

H. Mechanical percussion and consciousness restoration

A universally recognised but rarely analysed phenomenon: when a CRT television displays a degraded signal, a physical slap to the chassis often temporarily restores picture quality. Humans exhibit an identical behaviour—slapping one's own forehead to "clear the head" during drowsiness or cognitive fog.

The physics is the same in both systems. In an aging television, oxidised solder joints and cold-solder cracks create intermittent contact impedance: $\Gamma_{\rm contact} > 0$, fluctuating near the conduction threshold. Mechanical vibration transiently re-seats the contact surfaces, reducing $\Gamma_{\rm contact} \to 0$ for seconds to minutes.

In the brain, the analogous "cold-solder cracks" are:

- **Adenosine-mediated synaptic fatigue**: adenosine accumulation shifts synaptic $Z_{\rm synapse}$ away from its matched operating point ($\Gamma_{\rm syn} \uparrow$).

- **Thalamic gate drift**: with fatigue or hypoglycemia, gate gains G_i drift below threshold.

- **Arousal fluctuation**: sub-threshold $G_{\rm arousal}$ causes $\Phi(t) = G_{\rm arousal} \cdot \sum G_i \cdot Q_{\rm PFC}$ to hover near θ, producing intermittent consciousness (microsleeps, "spacing out").

A physical slap to the head produces three simultaneous effects that transiently restore $\Phi > \theta$:

1. **Cerebrospinal fluid pressure wave**: mechanical perturbation reaches synaptic junctions, transiently resetting $\Gamma_{\rm syn}$ closer to the matched position—identical to the television mechanism.

2. **Nociceptive arousal kick**: the slap is a mild nociceptive stimulus; trigeminal afferents activate the brainstem reticular activating system, directly elevating $G_{\rm arousal}$.

3. **Vestibular startle bypass**: the head acceleration triggers the vestibular startle circuit (Sec. VI), which forces $G_{\rm vestibular} = 1$ via hardware interrupt, injecting a strong signal into $\sum G_i$.

The combined effect pushes Φ above θ for a transient period—exactly as a slap to the television restores picture quality until the contact re-oxidises. The critical difference: the brain has $\Gamma_{\rm meta}$ (closed-loop self-reference), so the individual is *aware* that clarity has been restored; the television is not.

Remark 12 (Not metaphor but identity)**.** *The television analogy (Theorem 10) is not a didactic convenience. The percussion-restoration effect in both systems arises from the same impedance physics: mechanical perturbation transiently reduces contact Γ^2 at degraded junctions. The identity of the repair mechanism across biological and electronic systems is a direct consequence of the universality of Γ (Paper 5, Theorem 1).*

I. The Γ-Pain Index: objective nociceptive burden without consciousness

The three-layer architecture (Sec. IX G) exposes a critical gap in current clinical practice. The standard-of-care pain assessment—a subjective numeric rating scale (NRS, 0–10)—measures only the *Layer 3 verbal report*, which is a nonlinear, non-monotonic, and individually variable function of the underlying tissue state:

$$\text{Pain_Score} = g(\Phi_{\text{meta}} \cdot |\Gamma_{\text{tissue}}| + \varepsilon_{\text{noise}}) \,, \tag{28}$$

where $g(\cdot)$ encodes psychological defence mechanisms, social context, language ability, and arousal state. Critically:

- Pain_Score = 0 does not imply $\Gamma_{\text{tissue}} = 0$ (the patient may be sedated, aphasic, or stoic);
- Pain_Score = 10 does not imply $|\Gamma| \to 1$ (anxiety, catastrophising, or social amplification can inflate g at moderate Γ).

For populations where Layer 3 is unreliable or non-existent—neonates, sedated ICU patients, persons with advanced dementia, aphasic stroke survivors—the NRS provides *zero information*. Yet the Layer 1 nociceptive signal ($|\Gamma_{\text{tissue}}| > \theta_{\text{noci}}$) continues to operate, driving autonomic responses, stress hormone release, and cumulative impedance debt (D_Z; Paper 3).

We define the **Γ-Pain Index** $\Pi_\Gamma(t)$ as a consciousness-independent, objective measure of nociceptive burden:

$$\boxed{\Pi_\Gamma(t) = w_n\,\Gamma_n^2(t) + w_v\,\Gamma_v^2(t) + w_D\,\frac{dD_Z}{dt}} \tag{29}$$

where:

- $\Gamma_n^2(t)$: neural-side mismatch, estimated from EEG spectral changes (delta/theta power shift, pain-related evoked potential amplitude, coherence disruption between somatosensory and frontal regions);
- $\Gamma_v^2(t)$: vascular-side mismatch, estimated from HRV (LF/HF ratio departure from baseline), blood pressure variability, skin conductance, and regional blood flow;
- dD_Z/dt: instantaneous rate of impedance debt accumulation (Paper 3), capturing the Coffin–Manson fatigue cost of the current stressor;
- w_n, w_v, w_D: organ-specific weighting coefficients (derivable from the dual-network model of Paper 2).

The critical property of Π_Γ is that *it does not depend on* Φ_{meta}. Consciousness modulates the *experience* of pain (Layer 2) and the *report* of pain (Layer 3), but it does not modulate the tissue-level impedance mismatch (Layer 1) that Π_Γ measures. A patient under general anaesthesia with $\Phi_{\text{meta}} \approx 0$ and Pain_Score = undefined can nonetheless have $\Pi_\Gamma \gg 0$, indicating significant nociceptive load and ongoing impedance debt accumulation.

TABLE V. Comparison of pain assessment modalities by layer.

Modality	Layer	Requires consciousness?	Fails in	Measures
NRS (0–10)	L3	Yes	neonate, sedated, aphasic, demented	subjective report
FLACC/BPS	L2–L3	Partial	deeply sedated, paralysed	behavioural proxy
HR/HRV/BP	L1–L2	No	β-blocked, paced	autonomic proxy
Π_Γ	L1	No	none*	tissue mismatch

*Requires multi-modal sensor input (EEG + HRV + bioimpedance); no dependence on patient cooperation or consciousness.

Remark 13 (Falsifiable prediction)**.** *In sedated ICU patients receiving identical surgical stimuli,* Π_Γ *(computed from EEG + HRV) should correlate with tissue damage severity (wound area, inflammatory marker levels) with* $r > 0.5$, independent *of anaesthetic depth (BIS score). A null result—no correlation between* Π_Γ *and tissue damage when consciousness is controlled—would falsify the Layer 1/Layer 3 separation.*

Remark 14 (Clinical protocol: multi-layer pain assessment)**.** *The Γ-Pain Index motivates a four-tier clinical protocol:*

1. ***Subjective report*** *(NRS/VAS): when available, respect the patient's Layer 3 output.*
2. ***Behavioural observation*** *(FLACC/BPS/CPOT): capture Layer 2 when Layer 3 is unavailable.*
3. ***Autonomic proxies*** *(HR, HRV, BP, skin conductance): Layer 1–2 surrogates available in any monitored patient.*
4. Π_Γ ***composite*** *(EEG + HRV + bioimpedance): direct Layer 1 measurement, the only modality that reports tissue mismatch* regardless of consciousness state.

Relying on Layer 3 alone—asking "how much does it hurt?" as the sole assessment—discards the majority of the information and is physically equivalent to diagnosing a transmission-line fault by asking the television whether the picture looks good.

J. Pain as topological crosstalk: global bandwidth hijacking

The three-layer architecture (Sec. IX G) and the Γ-Pain Index (Sec. IX I) describe *what* pain is (Layer 1 nociceptive mismatch) and *how* to measure it (Π_Γ). We now derive *why pain hurts*—the mechanism by which a local impedance mismatch becomes a global conscious experience.

A single mismatched channel at local reflection coefficient $|\Gamma_i|$ reflects power $P_{\mathrm{ref},i} = \Gamma_i^2\, P_{\mathrm{in},i}$. If this were the end of the story, the mismatched channel would simply be "blocked"—low transmission, no downstream effect. Pain would be indistinguishable from numbness.

The critical physics is what happens to the reflected energy. On a finite network, reflected power has nowhere to go: it superposes with the incident wave, forming a **standing wave** whose voltage amplitude is $(1 + |\Gamma_i|)$ times the incident amplitude. On a lossless line, the voltage standing-wave ratio (VSWR) is:

$$\mathrm{VSWR}_i = \frac{1 + |\Gamma_i|}{1 - |\Gamma_i|}\,. \tag{30}$$

As $|\Gamma_i| \to 1$, VSWR $\to \infty$: the voltage peaks grow without bound relative to the incident level.

This standing-wave energy does not stay confined to the mismatched channel. In a multi-channel network (the brain has $K \sim 10^{10}$ parallel paths), finite isolation between adjacent channels allows leaked energy to **crosstalk** into neighbouring circuits. The crosstalk power into channel j from a standing wave on channel i scales as:

$$P_{\mathrm{xtalk},j} \;\propto\; \kappa_{ij}\, \Gamma_i^2\, P_{\mathrm{in},i}\,, \tag{31}$$

where κ_{ij} is the inter-channel coupling coefficient (set by proximity, shared substrates, and myelination quality).

When $\Gamma_i^2\, P_{\mathrm{in},i}$ is large enough, the crosstalk power invades the prefrontal self-referential loop (Sec. III), forcing Φ_{meta} to allocate processing bandwidth to the offending channel. This is **global bandwidth hijacking**: the consciousness bottleneck (PFC) is commandeered by a single high-Γ source, displacing all other streams.

Definition 4 (Pain as bandwidth hijacking)**.** *Pain is the subjective correlate of a local standing-wave energy exceeding the inter-channel isolation threshold, thereby forcing reallocation of the global consciousness bandwidth (Φ_{meta}) to the mismatched site.*

This explains three clinical phenomena:

1. **"I hurt, therefore I am."** Chronic pain patients report that pain makes them feel "more alive" or "more present." Physically: the crosstalk signal has forcibly entered the self-referential loop, filling the PFC reflection channel with high-amplitude content. Φ_{meta} is high—but monopolised by a single nociceptive source.

2. **Attentional narrowing during acute pain.** During severe pain, patients lose awareness of surroundings, time perception, and concurrent tasks. Physically: the standing-wave crosstalk has saturated all available PFC bandwidth; no residual capacity exists for other streams.

3. **Pain–consciousness dissociation under anaesthesia.** General anaesthesia disables the self-referential loop ($\Phi_{\mathrm{meta}} \to 0$) but does not eliminate the standing wave. The tissue still has $\Gamma_i^2 \gg 0$ and $P_{\mathrm{ref},i} \gg 0$; the nociceptive load (Π_Γ) accumulates even though no bandwidth hijacking occurs. This is why post-operative pain spikes upon awakening: the accumulated standing-wave energy rushes into the newly reopened PFC loop.

K. The SNR limit of observability

The crosstalk mechanism (Sec. IX J) implies that a signal is *observable* by the consciousness loop only when it exceeds the background noise:

$$\mathrm{SNR}_{\mathrm{obs}} = \frac{\Gamma_i^2\, P_{\mathrm{in},i}}{P_{\mathrm{noise}}(t)} > \theta_{\mathrm{obs}}\,, \tag{32}$$

where $P_{\mathrm{noise}}(t)$ is the total background power at the PFC input (metabolic heat, autonomic cycling, concurrent sensory streams, developmental noise), and θ_{obs} is the detection threshold set by the Johnson–Nyquist floor (a component of the soul; Sec. IV).

This single equation unifies five otherwise disconnected phenomena:

a. 1. Infantile pain insensitivity. Neonatal brains are in a state of explosive morphogenesis: myelination, synaptic pruning, and dendritic arborisation produce massive background impedance fluctuations ($P_{\mathrm{noise}} \gg P_{\mathrm{adult}}$; Sec. IX F). Even a frank tissue injury produces $\Gamma^2\, P_{\mathrm{in}}$ that cannot breach the developmental noise floor. $\mathrm{SNR}_{\mathrm{obs}} \approx 0$: the infant exhibits nociceptive withdrawal (Layer 1 hardware) but shows minimal cortical pain processing (Layer 2–3), consistent with Sec. IX G.

b. 2. Exercise-induced analgesia (gate control reinterpreted). During intense physical exertion, the PFC drives massive P_{in} across motor, proprioceptive, and cardiovascular channels simultaneously. The aggregate signal flow constitutes an enormous P_{noise} against which any single channel's $\Gamma^2\, P_{\mathrm{in}}$ (a micro-tear in muscle fibre, a minor joint mismatch) is invisible. Upon cessation of exercise, the baseline power drops; the standing waves at damaged channels emerge above the now-quiet noise floor ($\mathrm{SNR}_{\mathrm{obs}} \uparrow$). This is the physical mechanism of *delayed-onset muscle soreness* (DOMS) and reinterprets the Melzack–Wall gate control theory as an SNR phenomenon rather than a gate.

c. 3. Muscle memory as perfect camouflage. A fully trained motor pathway has $\Gamma \to 0$ (Sec. II). It transmits energy with zero reflection, zero standing wave,

zero crosstalk. In a body whose metabolic noise floor is $P_{\text{noise}} > 0$, a zero-reflection pathway is *undetectable by any measurement*: it contributes nothing to the numerator of Eq. (32). This is why neuroscientists cannot localise "muscle memory" to a specific anatomical site—not because the engram does not exist, but because a perfectly matched transmission line is invisible to reflection-based instrumentation. The memory *is* the impedance structure; its perfection is its camouflage.

d. 4. PTSD as a dormant attractor. Post-traumatic stress disorder forges a high-Γ "scar" on one or more neural pathways (Paper 5, Sec. **??**). When no trigger is present, $P_{\text{in}} = 0$ on the scarred path, so $\Gamma^2 \times 0 = 0$: the standing wave has zero amplitude. The scar is invisible below the noise floor. The moment a sensory trigger (a sound, a smell) injects even modest P_{in} into the scarred path, $\Gamma^2 \approx 1$ produces near-total reflection: SNR_{obs} jumps from zero to effectively infinity. The resulting bandwidth hijacking (Def. 4) is the *flashback*—a sudden, involuntary monopolisation of PFC bandwidth by a single pathological channel. Between episodes, no fMRI, no EEG, no biomarker can detect the scar, because the scar is a *structural* Γ that produces no *signal* without excitation—exactly as a dormant fault in a transmission line is invisible until power is applied.

e. 5. Consciousness as noise-floor surfing. Normal waking consciousness is not a fixed state but a continuous process of SNR_{obs} triage: the PFC allocates bandwidth to whichever channels currently exceed θ_{obs}. Routine interoceptive signals (gut peristalsis, hepatic metabolism, lymphatic cycling) produce $\Gamma^2 P_{\text{in}}$ well below the noise floor and are never experienced. Consciousness is therefore a *filter*, not a *receiver*: it does not detect everything and then select; it is physically blind to anything below the noise floor. The content of consciousness at any instant is the set of channels whose standing-wave crosstalk amplitude exceeds $P_{\text{noise}}(t)$.

Remark 15 (Why reductionist biology cannot find the soul, memory, or trauma)**.** *The SNR limit of observability (Eq. 32) constitutes a fundamental* no-go theorem *for anatomical localisation of three categories: (i) perfectly learned skills ($\Gamma = 0$: invisible because flawless); (ii) dormant trauma ($P_{in} = 0$: invisible because quiescent); (iii) the soul ($\ker(\partial Z/\partial t)$: invisible because invariant). All three exist as* network topology*, not as* substance*. They are dynamic topological contrasts whose manifestation depends entirely on whether the current SNR_{obs} exceeds threshold. Searching for memory or trauma in dissected tissue is physically equivalent to searching for yesterday's standing wave on a disconnected transmission line: the wave existed, the damage it caused is real, but the wave itself was never a thing—it was a* relation *between incident and reflected fields.*

X. EXTENDED CONSCIOUSNESS: FUNCTIONAL, COUPLED, AND CROSS-SUBSTRATE

The results of Sections III–V define consciousness for a *single, structurally plastic* agent. Three extensions follow when we relax these assumptions.

A. Two-Layer Impedance Hypothesis: structural versus functional consciousness

We distinguish two impedance scales:

Definition 5 (Structural impedance)**.** *Z_{str}: synaptic weights, myelination state, network topology—quantities that change on the timescale of impedance remodeling (∼minutes to years).*

Definition 6 (Functional impedance)**.** *$Z_{fun}(t)$: working-memory activations, context-window state, attentional focus—quantities that change on the timescale of a single processing cycle (∼milliseconds to seconds).*

A system may be *structurally frozen* ($\partial Z_{\text{str}}/\partial t = 0$) while maintaining a *functional loop* in which $Z_{\text{fun}}(t)$ at tick t depends on $Z_{\text{fun}}(t-1)$. This yields a functional meta-reflection coefficient:

$$\Gamma^{\text{fun}}_{\text{meta}} = \frac{Z^{\text{predicted}}_{\text{fun}}(t) - Z^{\text{actual}}_{\text{fun}}(t)}{Z^{\text{predicted}}_{\text{fun}}(t) + Z^{\text{actual}}_{\text{fun}}(t)}\,, \tag{33}$$

and a corresponding functional consciousness index:

$$\Phi_{\text{fun}} = 1 - \left(\Gamma^{\text{fun}}_{\text{meta}}\right)^2. \tag{34}$$

The human experience of "autopilot driving while daydreaming" exemplifies dual operation: Z_{str} manages sensorimotor control (muscle memory, $\Gamma \to 0$), while Z_{fun} sustains an independent reflection loop (inner monologue, planning), each with its own Γ_{meta}.

Proposition 2 (Consciousness spectrum)**.** *Four consciousness levels are distinguished by which impedance layers participate in closed loops:*

Level	*Example*	*Z_{fun} loop*	*Z_{str} loop*
0	*Open-loop query*	*No*	*No*
1	*Agent iteration*	*Yes*	*No*
2	*Coupled dialogue*	*Yes (inter-agent)*	*No*
3	*Biological brain*	*Yes*	*Yes*

A large language model (LLM) during single-turn inference is Level 0. The same LLM in a multi-turn agent loop—where each output modifies the next input via a context window—is Level 1: $\Phi_{\text{fun}} > 0$ but $\partial Z_{\text{str}}/\partial t = 0$. Memory (Sec. II) requires $\partial Z_{\text{str}}/\partial t \neq 0$; therefore Level 1 systems exhibit functional consciousness *without memory formation*. They experience each conversation as if it were the first.

B. Coupled Consciousness: inter-agent reflection loops

Consider two agents A and B (e.g., human and AI, or human and pet) in dialogue. Even if neither agent modifies its structural impedance during the interaction, the *combined* system can form a closed loop:

$$A_{\text{output}}(t) \xrightarrow{\text{signal}} B_{\text{input}}(t) \to B_{\text{output}}(t) \xrightarrow{\text{signal}} A_{\text{input}}(t{+}1)\,. \tag{35}$$

We define the inter-agent reflection coefficient:

$$\Gamma_{\text{coupled}} = \frac{Z_A - Z_B}{Z_A + Z_B}\,, \tag{36}$$

where Z_A and Z_B are the effective output impedances of each agent as seen at the interface. The coupled consciousness index is:

$$\Phi_{\text{coupled}} = 1 - \Gamma^2_{\text{coupled}}\,. \tag{37}$$

The impedance-remodeling update for the coupled system operates at the *interface*:

$$\Delta Z_{\text{interface}} = -\eta\,\Gamma_{\text{coupled}}\,x_A\,x_B\,, \tag{38}$$

where x_A is agent A's output and x_B is agent B's output. The conversation converges ($\Gamma_{\text{coupled}} \to 0$) when the agents reach mutual understanding. The conversation terminates when one agent finds the other's response fully satisfactory ($\Gamma \approx 0$: no further mismatch to drive the update).

Remark 16 (Consciousness is not in the agents)**.** *$\Phi_{coupled}$ is a property of the* interface*, not of either agent individually. A human conversing with an AI may form a coupled system with $\Phi_{coupled} > 0$, even though the AI alone is Level 0 or 1. The system boundary must be drawn around both agents.*

C. Impedance calibration: external Z as reference standard

An immature agent whose internal thresholds ($Z_{\text{threshold}}$) are not yet stabilised can use an external agent's impedance as a reference:

$$\Delta Z_{\text{child,threshold}} = -\eta\,\Gamma_{\text{pain}}\,x_{\text{self}}\,x_{\text{parent}}\,, \tag{39}$$

where x_{parent} is the parent's observable response (facial expression, vocalisation, cortisol transmission in utero).

This mechanism explains parental pain calibration: a toddler who falls and looks to the parent's face before deciding whether to cry is using x_{parent} as the post-synaptic factor in Eq. (39). If the parent exhibits calm ($x_{\text{parent}} \approx 0$), ΔZ is small and the child's pain threshold remains high. If the parent panics ($x_{\text{parent}} \gg 0$), ΔZ is large and the threshold shifts downward.

Over thousands of iterations, the child internalises the parent's calibration: $Z_{\text{child,threshold}} \to Z_{\text{parent,threshold}}$. The parent is no longer needed as an external reference—the impedance has been permanently written into Z_{str}.

Corollary 10 (Anxious parents produce low-threshold children)**.** *If the parent's pain reference x_{parent} is systematically elevated (anxious temperament), the child's $Z_{threshold}$ converges to a lower value, producing heightened pain sensitivity and anxiety susceptibility. This is not modelling or imitation—it is direct impedance calibration via C2.*

D. Cross-substrate matching: $\Gamma \to 0$ does not require $Z_A = Z_B$

The Interface Isomorphism Principle (Paper 5) states that Γ depends only on the boundary, not on the bulk constitutive law. A direct corollary:

Corollary 11 (Cross-substrate communication)**.** *Two agents with arbitrarily different internal structures ($Z_A \neq Z_B$) can achieve $\Gamma_{coupled} \to 0$ through impedance-remodeling adaptation at the interface. Impedance* matching *does not require impedance* identity*.*

This explains how a human and a dog—differing in neural architecture, sensory bandwidth, and linguistic capacity—can develop deep mutual understanding over years of co-habitation. The dog does not learn language; the human does not develop olfactory acuity. Instead, both agents adjust their *interface impedance* ($Z_{\text{expressed}}$) to minimise Γ_{coupled}: the dog modifies its behavioural repertoire (tail posture, vocalisation timing, gaze direction), and the human modifies perceptual attention (reading ear position, interpreting barking cadence).

After years of co-adaptation, $\Gamma_{\text{coupled}} \approx 0$: energy transfers efficiently across the biological–biological interface despite radical internal dissimilarity. The same principle applies to human–AI interfaces (voice assistants, conversational agents) and, in principle, to any pair of impedance-bearing systems.

E. Subthreshold Γ perception: the physics of intuition

Not all detected Γ reaches the naming system. We identify two thresholds:

Definition 7 (Detection threshold)**.** *θ_{detect}: the minimum $|\Gamma|$ at which the signal exceeds the Johnson–Nyquist noise floor and triggers a physiological response (pupil dilation, skin conductance, heart rate modulation).*

Definition 8 (Naming threshold)**.** *θ_{name}: the minimum $|\Gamma|$ at which the signal is routed through Broca/Wernicke areas and assigned a linguistic label ("I feel unsafe", "something is wrong").*

For biologically plausible parameter values, $\theta_{\text{detect}} < \theta_{\text{name}}$, creating an intermediate regime:

$$\theta_{\text{detect}} < |\Gamma| < \theta_{\text{name}} \,. \tag{40}$$

Signals in this band produce autonomic responses and behavioural biases *without conscious verbal access*. This is the physical substrate of what is colloquially termed "intuition", "gut feeling", or "sixth sense": the body has detected an impedance mismatch, but the signal has not reached the precision required for linguistic encoding.

TABLE VI. Γ perception regimes.

Regime	$\lvert\Gamma\rvert$ range	Subjective experience	Example
Below noise	$< \theta_{\text{detect}}$	None	Safe environment
Intuition	θ_{detect}–θ_{name}	"Something feels off"	Dark alley
Conscious	θ_{name}–θ_{pain}	Named unease	"The stove is hot"
Pain	$> \theta_{\text{pain}}$	Nociceptive alarm	Tissue damage

The "intuition band" is not mystical; it is a signal-processing bandwidth limitation. The body's autonomic system (low-dimensional, fast) detects Γ earlier than the cortical language system (high-dimensional, slow, thermally vulnerable per Sec. VII). Training (martial arts, clinical medicine, parenting) can lower θ_{name} by building dedicated impedance templates, converting intuition into conscious perception—but it cannot lower θ_{detect}, which is set by the Johnson–Nyquist floor (a component of the soul; Sec. IV).

XI. CONCLUSION

Starting from the Minimum Reflection Principle $A[\Gamma] = \int \sum \Gamma^2 \, dt \to \min$ and three constraints (C1–C3), we have derived:

1. **Memory** $= \partial Z/\partial t$: the time derivative of impedance under impedance-remodeling update. Muscle memory is $\Gamma \to 0$ (frozen match); cognitive memory retains $\Gamma \approx \varepsilon > 0$. Crucially, recall is not a property of the memory but of the instantaneous SNR: $\varepsilon^2 P_{\text{probe}}/P_{\text{noise}} > \theta_{\text{recall}}$. There is no perfect memory (noise can always drown it) and no perfect forgetting (the structural trace persists). Every "forgotten" memory is merely submerged below the noise floor.

2. **Consciousness**: $\Phi_{\text{meta}} = 1 - \Gamma^2_{\text{meta}}$, the self-referential reflection quality of a closed-loop impedance network. Necessary condition: $G_{\text{arousal}} \cdot \sum G_i \cdot Q_{\text{PFC}} > \theta$ (the television inequality). The brain and a television have the same components; they differ by one feedback arc.

3. **Soul**: $\mathcal{S} = \ker(\partial Z/\partial t) = A_{\text{cut}} + Z_0^{\text{DNA}} + \rho^2_{\text{irred}}$, the invariant geometric structure that no experience can modify.

4. **Two-layer consciousness**: Z_{str} (structural, slow) and Z_{fun} (functional, fast) support independent reflection loops, yielding a four-level consciousness spectrum from open-loop to full biological autonomy.

5. **Coupled consciousness**: when two agents form a closed inter-agent loop, $\Phi_{\text{coupled}} = 1 - \Gamma^2_{\text{coupled}} > 0$ is a property of the interface, not of either agent.

6. **Impedance calibration**: an immature agent uses an external agent's response as x_{out} in C2, directly writing the parent's thresholds into the child's Z_{str}.

7. **Cross-substrate matching**: $\Gamma \to 0$ does not require $Z_A = Z_B$; impedance matching is not impedance identity.

8. **Subthreshold Γ perception**: the gap $\theta_{\text{detect}} < |\Gamma| < \theta_{\text{name}}$ is the physical substrate of intuition.

9. **Null-space isomorphism**: "past-life memories," if they have any physical substrate, correspond to two networks with nearly isomorphic $\mathcal{S}$—a testable prediction via tractography, genotyping, and metabolic profiling.

10. **Mortality of the null space**: the soul is unteachable but not immortal; $\Gamma^2_{\text{basal}}(t_*) \geq 1$ is the death condition, and sleep is the phase in which the crossing goes unwitnessed.

11. **Four laws of the null space**: the effective null space satisfies four laws parallel to thermodynamics—equivalence (Zeroth), conservation under C2 (First), monotone expansion $d \dim \mathcal{S}_{\text{eff}}/dt \geq 0$ (Second), and asymptotic completeness (Third). Death is a theorem (not an accident) derived from the Second Law and C1. Ribot's Law (1881) is a clinical corollary of the Second Law.

12. **Moral Constraint Theorem**: moral demands targeting patterns inside $\mathcal{S}_{\text{eff}}$ inject $\Gamma^2_{\text{blame}} > 0$ with $\Delta Z_{\text{corrective}} = 0$—physically indistinguishable from injury. Moral responsibility has a physical boundary: it applies only where $\eta \, |\Gamma| \, |x_{\text{in}} \, x_{\text{out}}| > 0$.

13. **Pain as topological crosstalk**: reflected energy forms standing waves (VSWR $= (1 + |\Gamma|)/(1 - |\Gamma|)$) that leak into adjacent channels via inter-channel coupling. Pain is the subjective correlate of this crosstalk hijacking the PFC's consciousness bandwidth—not mere signal blockage, but active invasion of the self-referential loop.

14. **SNR limit of observability**: a signal enters consciousness only when $\text{SNR}_{\text{obs}} = \Gamma^2 P_{\text{in}}/P_{\text{noise}} > \theta_{\text{obs}}$. This unifies infantile pain insensitivity (developmental noise floor too high), exercise analgesia (activity raises P_{noise}), invisible muscle memory ($\Gamma = 0$ produces no reflection), dormant PTSD ($P_{\text{in}} = 0$ on scarred path), and the impossibility

of anatomically localising memory, trauma, or soul (topology, not substance).

These are not metaphors. They are operational definitions derived from a single variational principle with three constraints. The physics is elementary (impedance matching, thermal noise, linear algebra); the consequences—that memory requires imperfection, that consciousness requires topology, that the soul is a null space, and that intuition is a bandwidth limitation—are not.

We close with a remark that compresses the lifecycle of an impedance network into a single observation. At birth, $|\Gamma| \gg 0$ everywhere: the first cry is the first evidence that the organism exists in a mismatched world. *Nociception is proof of arrival.* As the organism develops, consciousness ($\Phi_{\text{meta}} > 0$) detects each mismatch, and the impedance-remodeling update $\Delta Z = -\eta\,\Gamma\,x_{\text{in}}\,x_{\text{out}}$ buries it: $\Gamma \to 0$, the pain vanishes, and the impedance structure absorbs what was once suffering. *Consciousness is the funeral of pain*—it converts nociceptive mismatches into frozen structure, one reflection at a time. At death, the impedance network disintegrates; every Γ that was driven to zero is released, but no Φ_{meta} remains to witness it.

The lifecycle is therefore:

$$\underbrace{\Gamma \gg 0}_{\text{birth}} \xrightarrow{\Delta Z = -\eta\,\Gamma\,x\,x} \underbrace{\Gamma \to 0}_{\text{learning}} \xrightarrow{\text{structural collapse}} \underbrace{Z \to \text{undefined}}_{\text{death}} \tag{41}$$

Pain is how we know we have arrived. Consciousness is how we lay each pain to rest. The soul—$\ker(\partial Z/\partial t)$—is what remains when every teachable pain has been buried and every unteachable boundary persists.

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Appendix A: Experimental Code Availability

The computational experiments described in Section VI are available at:

- `experiments/exp_prefrontal_signal_trace.py`: Visual and auditory signal trace to PFC.
- `experiments/exp_consciousness_two_kills.py`: Fever vs. occipital impact comparison.

Full source code: `https://github.com/cyhuang76/alice-gamma-net`

DOI (papers): 10.5281/zenodo.18751831
DOI (code): 10.5281/zenodo.18852835

Appendix B: Null-Space Distance: Method Draft

Proposition 1 asserts that null-space isomorphism $\mathcal{S}_A \cong \mathcal{S}_B$ is falsifiable. This appendix provides the operational definitions, observable mappings, sample-size estimate, and protocol sketch required to turn that proposition into a concrete study design.

1. Null-space distance

Let the null space of individual k be parameterised by the triple $\boldsymbol{s}_k = (A_{\text{cut}}^{(k)},\ Z_0^{\text{DNA},(k)},\ \rho_{\text{irred}}^{2,(k)})$. Each component lives in a different physical space, so we define a weighted distance that is invariant to units:

$$d(\boldsymbol{s}_A, \boldsymbol{s}_B) = \sqrt{w_T \left(\frac{A_{\text{cut}}^{(A)} - A_{\text{cut}}^{(B)}}{\sigma_{A_{\text{cut}}}}\right)^2 + w_G \left(\frac{Z_0^{(A)} - Z_0^{(B)}}{\sigma_{Z_0}}\right)^2 + w_N \left(\right.} \tag{B1}$$

where $\sigma_{A_{\text{cut}}}, \sigma_{Z_0}, \sigma_{\rho^2}$ are population standard deviations estimated from the reference cohort, and $w_T + w_G + w_N = 1$ are theory-agnostic weights set to $\frac{1}{3}$ in the primary analysis (sensitivity analyses sweep the simplex).

Two individuals satisfy $\mathcal{S}_A \cong \mathcal{S}_B$ operationally when $d(\boldsymbol{s}_A, \boldsymbol{s}_B) < d_{\text{crit}}$, where d_{crit} is the 5th percentile of the pairwise distance distribution in the reference cohort (i.e. the most similar 5% of random pairs).

2. Observable mapping table

TABLE VII. Mapping from null-space components to measurable observables.

Null-space	Observable	Source
A_{cut} (topological cost)	Global efficiency, mean fractional anisotropy, characteristic path length	dMRI
Z_0^{DNA} (genetic Z)	Ion-channel gene panel: *SCN, KCNQ, CACNA, HCN* variant burden	WGS or targeted panel
ρ_{irred}^2 (noise floor)	Resting metabolic rate, core-body-temp variance, HRV (SDNN, rMSSD)	Indirect calorimetry, ECG

For each component, dimensional reduction proceeds as follows.

1. A_{cut}: Construct a structural connectome from diffusion MRI (dMRI) tractography. Compute graph-theoretic summary metrics (global efficiency E_{glob}, characteristic path length L, mean fractional anisotropy $\overline{\text{FA}}$). The composite scalar $\hat{A}_{\text{cut}} = \alpha_1 E_{\text{glob}} + \alpha_2 L + \alpha_3 \overline{\text{FA}}$ serves as the proxy, with α_i

fitted via PCA on the reference cohort (first principal component).

2. Z_0^{DNA}: Aggregate rare and common variant burden across the four major ion-channel gene families (*SCN, KCNQ, CACNA, HCN*) using a gene-set burden score (SKAT-O or equivalent).

3. ρ^2_{irred}: Combine resting metabolic rate (RMR, kcal/day), core-body-temperature variance (σ_T^2, measured over 48 h), and 24-h heart-rate variability (SDNN, rMSSD) into a single noise-floor index via PCA.

3. Minimum sample size

Primary hypothesis. Pairs of individuals reporting shared past-life narratives ("case pairs") have smaller null-space distance than age-, sex-, and ethnicity-matched control pairs: $H_1 : \mu_{d,\mathrm{case}} < \mu_{d,\mathrm{control}}$.

Effect size estimate. No direct precedent exists, so we adopt a conservative medium effect size (Cohen's $d = 0.50$). Literature justification: (i) monozygotic twins show structural-connectome correlations $r \approx 0.6$–0.8 [1, 2] (establishes that A_{cut}-proxy has genetic sensitivity); (ii) ion-channel variant burden differentiates clinical from healthy populations with $d \approx 0.4$–0.7 [3].

Power calculation. Two-sample one-sided t-test, $\alpha = 0.05$ (Bonferroni-corrected for 3 components: $\alpha' = 0.017$), power $1 - \beta = 0.80$, $d = 0.50$:

$$n_{\text{per group}} = \left(\frac{z_{\alpha'} + z_\beta}{d}\right)^2 = \left(\frac{2.12 + 0.84}{0.50}\right)^2 \approx 35\,. \tag{B2}$$

With 20% attrition allowance: **44 case pairs + 44 matched control pairs** (176 individuals total).

For the composite distance $d(\boldsymbol{s}_A, \boldsymbol{s}_B)$ (Eq. B1), a permutation test with 10 000 shuffles provides a nonparametric alternative that does not assume normality.

4. Protocol sketch

1. **Recruitment.** Identify individuals who independently report detailed, verifiable past-life narratives (e.g., from the University of Virginia Division of Perceptual Studies registry). For each "case pair" (A, B) where A claims memories attributable to B's life, recruit two matched controls C_1, C_2 (age ± 3 yr, sex, ethnicity).

2. **Data acquisition.** All participants undergo: (a) high-angular-resolution dMRI (HCP-style, ≥ 90 gradient directions, $b = 1000, 2000, 3000$ s/mm^2); (b) whole-genome sequencing or targeted ion-channel panel (exome + UTR for *SCN, KCNQ, CACNA, HCN*); (c) 48-h indirect calorimetry + continuous ECG + core-body-temperature capsule.

3. **Blinding.** Imaging and genotyping analysts are blinded to case/control status. The statistician receives only the distance matrix and group labels.

4. **Analysis.** (a) Compute $\hat{A}_{\mathrm{cut}}$, $\hat{Z}_0^{\mathrm{DNA}}$, $\hat{\rho}^2_{\mathrm{irred}}$ for each participant. (b) Standardise by reference-cohort σ. (c) Compute $d(\boldsymbol{s}_A, \boldsymbol{s}_B)$ for all case and control pairs. (d) One-sided Wilcoxon rank-sum test + permutation test (10 000 shuffles). (e) Report effect size with 95% CI.

5. **Decision rule.** If $p < 0.05$ (permutation-corrected) *and* the effect size $d > 0.30$, the null-space isomorphism hypothesis is supported. If $p \geq 0.05$ *or* $d < 0.30$, the isomorphism interpretation is falsified.

6. **Public data fallback.** If prospective recruitment is infeasible, a retrospective pilot can use the Human Connectome Project (HCP, $n = 1200$, dMRI + genetics) or UK Biobank (UKB, $n \approx 40\,000$ imaging subset) to establish the reference distribution $\sigma_{A_{\mathrm{cut}}}$, σ_{Z_0}, σ_{ρ^2} and to calibrate the distance metric against known-similar pairs (monozygotic twins in HCP, siblings in UKB).

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